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RESEARCH

An empirical framework for testing theories about complementarity in organizational design

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Abstract

This paper studies alternative empirical strategies for testing for complementarity in organizational design. We study a model where firms choose whether or not to adopt multiple organizational practices, and these choices interact to determine performance according to an "organizational design production function." Unobserved variation in the returns to individual practices influences both the decision to adopt practices and realized performance. As a result, common cross-sectional empirical approaches used to test for complementarity may be subject to bias. Our analysis (a) identifies specific and plausible conditions under which bias exists, (b) analyzes whether the sign of bias can be determined, and (c) provides interpretations in terms of the underlying economic and managerial environment. In certain cases, several different procedures all result in qualitatively similar biases. We further propose a structural estimation approach that includes simultaneous equations for productivity and the demand for organizational practices. Testing for complementarity requires choice-specific instrumental variables (i.e., observable factors impacting the adoption of individual practices but independent of other practices and measured performance) and an appropriate treatment for selectivity. Our analysis offers implications for data gathering, survey design, and the interpretation of empirical evidence concerning complementarity in organizational design.

JEL Classification L23 · D2 · C3 · C52

Prefatory note

The primary purpose of this paper is to develop an econometric framework designed to rigorously test for complementarities in organizational design practices. By leveraging a model where firms choose whether or not to adopt multiple organizational practices, and these choices interact to determine performance according to an "organizational design production function," we provide a nuanced understanding of the interaction effects between different organizational practices. Our empirical framework allows us to distinguish between different sources of bias that might arise from unobserved heterogeneity and offers a consistent test for complementarity, thereby providing more accurate insights into organizational behavior and performance outcomes.

Our contributions extend beyond theoretical advancements by addressing practical implications for organizational design and policy-making. In particular, we propose a structural estimation approach that accounts for unobserved heterogeneity and utilizes choice-specific instrumental variables to correct for potential biases. This approach not only refines the identification of complementarities but also enhances the precision of estimates in small-sample contexts where traditional methods might fall short. Our framework's flexibility and robustness make it particularly suitable for real-world data sets, such as those involving customer service centers or other service industries with heterogeneous economic environments. Thus, our paper offers both a methodological contribution and practical tools for improving organizational efficiency and decision-making.

Since the original drafting of this paper, a rich body of literature on organizational complementarities has

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emerged both building on and complementing our initial approach. Key contributions include Caroli and Van Reenen's (2001) examination of human capital and organizational change, Bresnahan et al.'s (2002) analysis of information technology and work organization, Cassiman and Veugelers's (2006) study of innovation strategy, Cassiman and Valentini's (2016) exploration of complementarities in corporate strategies, and Novak and Stern's (2008; 2009) investigations into vertical integration and organizational design in the automotive industry. Each of these works has significantly advanced our understanding of how different organizational practices interact to influence firm performance.

The industrial organization and marketing literatures have also considered the question of complementarity in the context of consumer demand. In an early application, Train et al. (1987) analyze demand for bundles of products, analogous to the "system" modeling in our paper. Gentzkow (2007) studies demand for news products, modeling complementarity among online and offline news sources using a parameter in consumers' utility. Song and Chintagunta (2007) analyze complementarity between supermarket purchases of laundry detergent and fabric softener. Berry et al. (2014) provide a review of the industrial organization literature on complementarity. More recently, Ruiz et al. (2020) introduce a model of complementarity among products along with an estimation approach that is computationally feasible for large numbers of products. Their "SHOPPER" model incorporates probabilistic elements to capture consumer choices among substitutes and complements, but also imposes restrictions on the unobservable parameters governing complementarity, reflecting the same principles of interaction and complementarity that our framework addresses within firms. The progression of this literature underscores the broader relevance of our work and its potential to inform various domains where understanding the interplay of multiple factors is crucial.

Introduction

Quantitative research in strategy and managerial economics has increasingly focused on the impact of organizational design choices on performance. While many of these studies consider a single element or dimension of organization design (e.g., focusing on the "returns" to the adoption of information technology in isolation), a growing literature emphasizes the potential importance of interactions between elements of organizational design. A common finding of the empirical literature is that organizational design practices are "clustered": the adoption of one practice is correlated with the adoption of other practices, and, as a result, "sets" of practices consistently appear together.[1]

A rich theoretical debate has arisen attempting to explain clustered organizational practices, with implications for adoption and performance. In particular, theoretical perspectives in both strategy and organizational economics suggest that interdependencies among practices can be crucial for determining the payoffs for individual practices (Thompson 1989; Milgrom and Roberts 1990; Levinthal 1997).[2] An important strand of the literature has emphasized the potential for complementarity, which is defined as positive interdependencies among pairs of practices (Milgrom and Roberts 1990; Holmstrom and Milgrom 1994; Siggelkow 2002). A surge of empirical work in recent years builds directly on these theoretical analyses, explicitly "testing" for complementarity using a variety of approaches.[3] However, most studies do not explicitly account for the potential impact of unobserved variation in the costs and benefits of organizational design practices on the interpretation of empirical evidence relating to tests for complementarity or interdependencies.

This paper develops a formal econometric framework designed to allow a more complete evaluation of why practices appear together and how joint adoption affects firm-level productivity. This model allows analysis of the sources of bias that may be present in many of the econometric approaches used in the literature, and the conditions under which bias can be signed and interpreted. We provide sufficient conditions for identification of the structural parameters of the "organizational design production function" and a consistent test for complementarity. The analysis is tailored for cross-sectional applications where many firms face similar production technologies, make comparable choices about organizational design, but face different costs or benefits to adoption. For example, retail service outlets (such as a customer service center) are designed to accomplish similar goals, but operate in economic environments differing in demographic characteristics, labor regulations,

[1] Early empirical work emphasizing the presence of clustering includes, among others, Anderson and Schmittlein (1984), Arora and Gambardella (1990), Helper and Levine (1994), and Macduffie (1995).

[2] Some researchers highlight interdependencies as a foundation for limited rationality on the part of organizations (Levinthal 1997; Rivkin and Siggelkow 2003).

[3] While a comprehensive review of this voluminous empirical literature is beyond the scope of this paper, studies examining the complementarity hypothesis include Slade (1996), Ichniowski et al. (1997), Ichniowski and Shaw (1999), Brickley (1999), Hwang and Weil (1996), Bresnahan et al. (2002), and Cassiman and Veugelers (2006). See Ichniowski and Shaw (2003) for a useful review of the evidence relating to complementary HR practices. As well, we (along with co-authors) have pursued several empirical studies along these lines, including Athey and Stern (1999), Athey and Stern (2002), Cockburn et al. (2003), and Novak and Stern (2008).

or technological infrastructure. These organizations make (potentially) interdependent choices about factors such as on-the-job training and the use of information technology (IT), which together determine overall performance.

Establishing a testing framework for complementarity is important, given the business and public policy implications of interdependent organizational design choice. For example, if a training subsidy affects the adoption of training programs, it will also have indirect effects on the adoption and productivity of complementary choices, such as the adoption of advanced IT.[4] Consequently, optimal subsidies would account for the direct and indirect effects on organizational design. As well, complementarity between a set of practices implies that the adoption of one practice raises the returns to adopting other practices within the firm; as a result, the presence of complementarity among practices raises the returns to coordinated decisionmaking in organizational practice adoption.

Interactions between elements of organizational design present several distinct empirical challenges that do not normally arise in the context of traditional economic analysis of productivity. First, in contrast to analyses of traditional factor inputs such as capital or labor, the econometrician does not typically observe the relevant "input prices" that each firm faces in adopting an organizational practice. For this reason, the tools of duality, which are exploited throughout the productivity literature in economics, are not applicable, and it is thus necessary to confront the difficulties associated with direct estimation of the production structure. In particular, we must allow for the possibility that the costs and benefits to employing practices might vary across firms, and yet be unobservable.

A second challenge arises from the fact that some sets of practices tend to be adopted in clusters (as theory would suggest, when practices are complements). The presence of clustering implies that some combinations of practices will occur only infrequently, and so it may be difficult to precisely estimate the parameters describing the interactions between these parameters using a regression of productivity on practice combinations. We accommodate this potential difficulty by exploiting revealed preference: the fact that firms have chosen to adopt practices together is potentially informative about the joint returns to the practices. However, the econometric procedures that rely on revealed preference require that we observe choice-specific instrumental variables (i..e, observable factors impacting the adoption of individual practices but independent of other practices and measured performance). Using such instrumental variables, a consistent test for complementarity is feasible.

Our formal analysis begins by introducing a model of an "organizational design production function," with parameters that specify the interactions between choices, as well as exogenous variables that determine the costs and benefits of each practice. The firm chooses a set of "organizational design practices" taking into account the relevant costs and benefits of adoption. We focus on a model of a cross-section of firms facing stable production parameters, and assume that both the firm and econometrician observes choice-specific instrumental variables. As well, to demonstrate potential sources of bias from traditional procedures, our analysis includes choice-specific variation observed by the firm but unobserved to the econometrician. This variation explains differences in firm practices that cannot be explained by observables but affect the marginal returns or costs of adoption.[5]

Building on results from the econometrics literature (e.g. Heckman and Honore (1990); Thompson (1989); Ichimura and Lee (1991)), this paper establishes conditions under which the parameters of the production function and the joint distribution of unobservables are identified. In the most general model (which we term the Random Systems Model (RSM)), each combination of practices, or "system," (for example, the combined adoption of training programs, job security, and incentive pay) is subject to a random shock.[6] In such a model, only the distribution of interactions among practices is identified; practices are complements for some firms and substitutes for others. Without additional structure, a unique test for complementarity is not feasible, and one cannot test whether practice adoption is consistent with optimization.

We then identify a restriction on the Random Systems Model, which we call the Random Practice Model (RPM), which allows for sharper business and public policy implications. The RPM essentially assumes that there is no unobserved variation in the interactions between practices. This is analogous to assuming a constant elasticity of substitution between inputs. In other words, the RPM allows for unobserved returns to individual practices, but assumes that unobserved returns to one practice do not depend on the adoption of other practices.

[4] Complementarity between IT and organizational practices enhancing worker skills may be particularly important, as skill-biased technical change has been suggested as a key determinant of increasing wage and income inequality (Autor et al. 1998; Bresnahan et al. 2002; Autor et al. 2003).

[5] In problems of internal organizational design, these variables correspond to factors such as the talent and past experiences of managers and workers, the beliefs held within the firm about current and future market conditions, labor-management relations, the formal and informal processes for adopting changes in organizational design in a given firm, the influence exerted by various interest groups within the firm, and other adjustment costs.

[6] The RSM model is an application of the general "switching regressions" model (Heckman and Macurdy 1986) where an agent chooses between several discrete choices (where each choice is an organizational design system).

We use this formal structure in several ways. We first focus on the conditions under which a set of empirical procedures used in the existing literature provides a consistent test for complementarity between a given pair of practices. To do so, we distinguish between two conditions: (TC) the practices are complements in the organizational design production function, and (TI) the practices are technologically independent. Previous empirical studies have attempted to distinguish between (TC) and (TI) in two main ways: first, by testing whether the practices are positively correlated, conditional on observables; and second, by using OLS or instrumental variables approaches to estimate the parameters of a productivity equation and test whether the interaction effect between practice pairs is positive.

Our analysis highlights how particular forms of unobserved heterogeneity bias these procedures in specific directions. Consider the following alternative assumptions about the unobserved returns: (a) the unobserved returns among practices are affiliated (a strong form of positive correlation) and (b) the unobserved returns are independent. Even when the choices do not interact in determining productivity (TI), the presence of positive correlation between the unobserved returns to the two different practices yields (i) positive correlation in adoption among practices and (ii) a positive estimate of the interaction effect in an OLS productivity regression. More generally, positive correlation in the unobservables results in a force for a positive bias in the estimate of interaction effects in a productivity regression.

In contrast, complementarity between practices (TC) results in a competing effect for the direction of bias: (TC) creates a force towards understating interaction effects. Under (TC), adopting a given practice (such as a training program) leads to a less favorable selection of firms adopting complementary practices (such as advanced IT). If the unobserved returns to practices are independent and (TC) holds, the bias on the interaction effect will be negative.

We then evaluate instrumental variables approaches. Similar to OLS, the "traditional" 2SLS estimate does not provide a consistent test for complementarity. However, we demonstrate that when there is a binary (dummy) instrumental variable for each practice, it is possible to construct a consistent test for the hypothesis of no interaction effects by calculating a test statistic based on Local Average Treatment Effects (along the lines of Imbens and Angrist (1994)); however, even this procedure fails to offer a consistent test for complementarity when complementarity is in fact present.

Next, we use the formal model to analyze the properties of a structural estimator of the parameters with two main features: (i) it explicitly models the distribution of the unobserved heterogeneity, and (ii) it includes a system of equations, including an equation describing productivity and a set of equations describing the practice adoption decisions. There are several advantages to such a structural approach. First, by accounting for the unobserved heterogeneity, it is possible to obtain consistent estimates of the parameters of the organizational design production function as well as the covariance between the unobserved returns to different organizational design practices. Second, our model nests all prior models we are aware of, and so direct comparisons can be made between the implicit assumptions associated with previous approaches. Third, by specifying an internally consistent simultaneous equations system, we can impose the cross-equation restrictions on the interaction effects; as suggested above, since organizational design practices are often positively correlated in applications, the use of revealed preference can yield substantial efficiency gains. Finally, (if the RPM is appropriate) we can perform tests about the process that leads to the adoption of practices, including whether or not practice adoption appears consistent with optimal behavior.

Overall, our analysis provides an integrated approach to testing for complementarity in organizational design. Similar to other applied researchers emphasizing the importance of endogeneity and selectivity in managerial research (see Hamilton and Nickerson (2003) for a particularly useful discussion), our model highlights the potential for bias using traditional approaches that do not account for unobserved heterogeneity. However, it is important to emphasize that the nature of the selection biases that arise from statistical and technological interactions between choices is subtle and has not received attention in the existing applied econometrics literature. A well-grounded test for complementarity requires a careful assessment of the source of econometric identification (e.g., by observing choice-specific instrumental variables) and an econometric framework that accounts for the specific forms of unobserved heterogeneity.

The paper proceeds as follows. Section "The model" presents the formal model of the organizational design production function. Section "Unobserved heterogeneity and reduced-form tests for complementarity" analyzes the implicit assumptions, potential biases and interpretations associated with prior approaches. Section "A structural approach to estimation and testing in organizational design" considers the identification and properties of a structural model. Section "Issues for data gathering and survey design" discusses issues for data collection and survey design; a final section concludes.

The model

This section develops a model for analyzing cross-sectional data on the adoption and productivity of organizational design practices. The model is general enough to

Table 1 Table of notation

Notation	Description	Observed/Unobserved endogenous/exogenous
Variables		
$y^t = (y_1^t, \ldots, y_J^t)$	Vector of J discrete choices made by firm t.	Observed; endogenous.
$\mathrm{x}^t = \left(\mathrm{x}^{0,t}, \mathrm{x}^{1,t}\right) = \left(\left(x_1^{0,t}, \ldots, x_J^{0,t}\right), \left(x_1^{1,t}, \ldots, x_J^{1,t}\right)\right)$	Vector of practice-specific exogenous variables which affects observable performance (f); $x_j^{0,t}$ is the return when $y_j^t = 0$, while $x_j^{1,t}$ is the return when $y_j^t = 1$.	Observed; exogenous.
$\chi^t = \left(\chi^{0,t}, \chi^{1,t}\right) = \left(\left(\chi_1^{0,t}, \ldots, \chi_J^{0,t}\right), \left(\chi_1^{1,t}, \ldots, \chi_J^{1,t}\right)\right)$	Similar to x.	Unobserved; exogenous.
$\mathrm{w}^t = \left(\mathrm{w}^{0,t}, \mathrm{w}^{1,t}\right) = \left(\left(w_1^{0,t}, \ldots, w_J^{0,t}\right), \left(w_1^{1,t}, \ldots, w_J^{1,t}\right)\right)$	Vector of practice-specific exogenous variables which does not affect performance, otherwise similar to x.	Observed; exogenous.
$\omega^t = \left(\boldsymbol{\omega}^{0,t}, \boldsymbol{\omega}^{1,t}\right) = \left(\left(\omega_1^{0,t}, \ldots, \omega_J^{0,t}\right), \left(\omega_1^{1,t}, \ldots, \omega_J^{1,t}\right)\right)$	Similar to w.	Unobserved; exogenous.
$\mathrm{z}^t = \left(z^t_{(0,\ldots,0)}, \ldots, z^t_{(1,\ldots,1)}\right)$	Vector of system-specific exogenous variables which affect productivity.	Observed; exogenous.
$\zeta^t = \left(\zeta^t_{(0,\ldots 0)}, \ldots, \zeta^t_{(1,\ldots 1)}\right)$	Vector of system-specific exogenous variables which affect productivity.	Unobserved; exogenous.
$\mathrm{u}^t = \left(u^t_{(0,\ldots,0)}, \ldots, u^t_{(1,\ldots 1)}\right)$	Vector of system-specific exogenous variables which do not affect productivity.	Observed; exogenous.
$\nu^t = \left(\nu^t_{(0,\ldots,0)}, \ldots, \nu^t_{(1,\ldots 1)}\right)$	Vector of system-specific exogenous variables which do not affect productivity.	Unobserved; exogenous.
Parameters		
θ	Parameter of function f which determines supermodularity of f.	To be estimated.
$\alpha, \eta, \beta, \delta$	Parameters which affect the returns to exogenous variables.	To be estimated.
Functions		
π	The firm's overall objective function.	No measure observed.
f	The firm's performance or output.	Potentially observed.
g	The portion of the firm objective which does not directly affect f.	No measure observed.
F	The joint distribution over the unobservables ζ.	Marginals to be estimated.
G	The joint distribution over the unobservables $\zeta + \nu$.	To be estimated.

incorporate alternative assumptions about the following elements: (i) the nature of interaction effects between practices in the "production function," (ii) the mechanism through which practice adoption decisions are determined, and (iii) the nature of the joint distribution of unobserved returns.

The model is tailored to applications where there are organizations with similar objectives and options operating in heterogeneous economic environments at a given point in time.[7] Consider a hypothetical telephone customer service center whose goal is to maximize the number of customers served by each call-taker, subject to a quality constraint. The firm makes choices about two practices that may improve productivity: the level of worker training and the adoption of advanced computing equipment. Costs and benefits to adoption may vary in observable and unobservable ways. For example, workers' expectations about their job tenure affect their return to making specific investments, thus influencing the firm's incentive to provide the training course. The benefits to computer technology may depend on the previous experience of managers and workers with similar technologies. In this context, we may wish to test whether computers and training are complementary. In order to conduct such a test, it will also be important that some factors that affect adoption, but not productivity in use, are observed. For example, some call centers might be in states where training is subsidized, and there might be variation in the pre-existing telecommunications infrastructure which affects the costs of adopting a system with caller identification or other advanced features.

We proceed by developing an abstract model of the organizational design production function and the firm's "demand" for organizational design practices. We then introduce

[7] This focus highlights the assumptions required for identification as well as the difficulties that arise from reduced-form analysis in a cross-sectional setting. The many interesting issues associated with panel data and the dynamics of practice adoption are left for future research; Athey and Stern (2002) provides a discussion of some of these issues in the context of the dynamics of adoption of E911 technology and the implementation of training programs.

econometric assumptions about observability and different types of unobserved heterogeneity and discuss the tradeoffs associated with restrictions on the stochastic structure.

The organizational design production function

This section introduces the organizational design production function and the restrictions on that function implied by complementarity among practices. The notation is summarized in Table 1. We consider a firm t where a vector of J practices, denoted $\mathrm{y}^t = (y_1^t, \ldots, y_J^t)$, is endogenously determined. We focus on the case where each of the practices y_j^t is a discrete choice from $\{0,1\}$, resulting in 2^J distinct combinations or organizational "systems." A training program would be considered a practice, while a system is described by the choices the firm makes about the whole vector of practices. When there are two practices to be adopted, $\mathrm{y}^t \in \{0,1\} \times \{0,1\}$. The "systems" in this model are the practice combinations $\mathrm{y}^t \in \{(1,1),(1,0),(0,1),(0,0)\}$.

The productivity of different combinations of organizational design elements varies across firms from two sources: practice-specific and system-specific. First, system-specific exogenous variables, denoted $\mathrm{Z}^t = (Z_1^t, \ldots, Z_K^t)$, change the returns to the joint adoption of practice combinations. On the other hand, practice-specific exogenous variables, denoted

$\mathrm{X}^t = (X_1^t, \ldots, X_J^t)$, affect the incremental gain in productivity from adopting a practice irrespective of what other elements are adopted.

Productivity, denoted f^t, is determined as a function of these variables, $f^t = f\left(\mathrm{y}^t, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right)$, where $\mathrm{M} = (\boldsymbol{\theta}, \boldsymbol{\alpha}, \boldsymbol{\beta})$. The system-specific return to system $\mathrm{y} \in \{0,1\}^J$ is $\theta_{\mathrm{y}} + \alpha_{\mathrm{y}} Z_{\mathrm{y}}^t$, while the practice-specific payoff for practice j is $\beta_j^0 X_j^{0,t}$ if $y_j^t = 0$, $\beta_j^1 X_j^{1,t}$ if $y_j^t = 1$. For the two-choice example, then, the functional form for productivity is:

$$
\begin{aligned}
f\left(\mathbf{y}^t, \mathbf{X}^t, \mathbf{Z}^t; \mathbf{M}\right) = {} & \left(1-y_1^t\right)\left(1-y_2^t\right) \cdot \left[\theta_{00} + \alpha_{00} Z_{00}^t\right] \\
& + \left(1-y_1^t\right) y_2^t \left[\theta_{01} + \alpha_{01} Z_{01}^t\right] \\
& + y_1^t \left(1-y_2^t\right) \cdot \left[\theta_{10} + \alpha_{10} Z_{10}^t\right] \\
& + y_1^t y_2^t \cdot \left[\theta_{11} + \alpha_{11} Z_{11}^t\right] \\
& + \left(1-y_1^t\right) X_1^{0,t} \beta_1^0 + y_1^t X_1^{1,t} \beta_1^1 \\
& + \left(1-y_2^t\right) X_2^{0,t} \beta_2^0 + y_2^t X_2^{1,t} \beta_2^1
\end{aligned}
\tag{1}
$$

Equation (1) highlights several key features of the model. First, the model allows for interactions between the practices: the productivity of one practice depends on the adoption decisions about the other practices. Second, the productivity of individual practices varies across firms, as formalized in the interaction between X and y. Third, complementarity among practices is a parametric restriction on the organizational design production function. Recall a formal definition for complementarity and supermodularity:

Definition Fix $\mathrm{X}^t, \mathrm{Z}^t$, and M. Two practices y_i and y_j are complements in the objective function f if the following inequality holds for all y_{-ij}^t:

$$
\begin{aligned}
& f\left(y_i^H, y_j^H, \mathrm{y}_{-ij}, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right) - f\left(y_i^L, y_j^H, \mathrm{y}_{-ij}, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right) \\
& \quad \geq f\left(y_i^H, y_j^L, \mathrm{y}_{-ij}, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right) - f\left(y_i^L, y_j^L, \mathrm{y}_{-ij}, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right)
\end{aligned}
\tag{2}
$$

f is supermodular in y if y_i and y_j are complements for all $i \neq j$.

It is also possible to define complementarity conditional on a particular set of choices about the other practices, y_{-ij}^t. When there are only two choices (1 and 2), then these choices are complements if

$$
\begin{aligned}
\kappa_{12} = {} & \left(\left[\theta_{11} + \alpha_{11} Z_{11}^t\right] - \left[\theta_{01} + \alpha_{01} Z_{01}^t\right]\right) \\
& - \left(\left[\theta_{10} + \alpha_{10} Z_{10}^t\right] - \left[\theta_{00} + \alpha_{00} Z_{00}^t\right]\right) \geq 0
\end{aligned}
\tag{3}
$$

If Z^t varies across firms, then different firms will have different levels of complementarity between practices. However, in the absence of Z^t, the interaction effects are fixed across firms and (3) reduces to:

$$
\kappa_{12} = (\theta_{11} - \theta_{01}) - (\theta_{10} - \theta_{00}) \geq 0 \tag{4}
$$

Intuitively, (4) says that the returns to adopting practice 1 are higher when practice 2 has been adopted.[8] Finally, it is sometimes useful to rewrite (1) in a way that highlights its analogy to a switching regressions model (Heckman and Macurdy 1986):

$$
f\left(\mathrm{y}^t, \mathrm{X}^t, \mathrm{Z}^t; \mathrm{M}\right) = \begin{cases}
\theta_{00} + Z_{00}^t \alpha_{00} + X_1^{0,t} \beta_1^0 + X_2^{0,t} \beta_2^0 & \text{if } \mathrm{y}^t = (0,0) \\
\theta_{01} + Z_{01}^t \alpha_{01} + X_1^{0,t} \beta_1^0 + X_2^{1,t} \beta_2^1 & \text{if } \mathrm{y}^t = (0,1) \\
\theta_{10} + Z_{10}^t \alpha_{10} + X_1^{1,t} \beta_1^1 + X_2^{0,t} \beta_2^0 & \text{if } \mathrm{y}^t = (1,0) \\
\theta_{11} + Z_{11}^t \alpha_{11} + X_1^{1,t} \beta_1^1 + X_2^{1,t} \beta_2^1 & \text{if } \mathrm{y}^t = (1,1)
\end{cases}
\tag{1'}
$$

The switching regressions model (1') takes the "system" as the unit of analysis, while the production function model (1) emphasizes that each system is composed of a set of practices, each of which comes with a separate set of benefits and costs.

Demand for organizational design practices

The mapping between the exogenous variables and firm's choices about practice adoption describes the firm's "demand"

[8] When there are multiple choices, there will be a distinct inequality corresponding to each combination of choices about the other practices.

for these practices and so is, to some extent, analogous to the factor demand equation used throughout productivity economics. Since X^t and Z^t affect the returns to each of the practices, demand clearly depends on these variables.[9] As well, there may be factors which affect adoption but not measured productivity (such as regulation or the presence of subsidies), and so we incorporate practice-specific exogenous variables, $\mathrm{W}^t = (W_1^t, \ldots, W_J^t)$, and system-specific variables, $\mathrm{U}^t = (U_1^t, \ldots, U_K^t)$. The demand for a given practice is thus determined according to $y_j^t = D_j\left(\mathrm{X}^t, \mathrm{Z}^t, \mathrm{W}^t, \mathrm{U}^t; \mathbf{\Lambda}\right)$ (where Λ is a vector of parameters of the demand equation), and a policy of optimal adoption implies:

$$\begin{aligned}\mathbf{y}^* &= \arg\max_{\mathbf{y}} \pi\left(\mathbf{y}^t, \mathbf{X}^t, \mathbf{Z}^t, \mathbf{W}^t, \mathbf{U}^t; \mathbf{M}, \mathbf{N}\right) \\ &\equiv f\left(\mathbf{y}^t, \mathbf{X}^t, \mathbf{Z}^t; \mathbf{M}\right) + g\left(\mathbf{y}^t, \mathbf{W}^t, \mathbf{U}^t; \mathbf{N}\right)\end{aligned} \tag{OPT}$$

(OPT) highlights two issues for our analysis. First, π represents the objective function faced by the agent responsible for decision-making, and (OPT) assumes that this agent efficiently chooses organizational design practices in a manner consistent with (at least) their own idiosyncratic incentives and perhaps the overall (long-run) profits of the organization.[10] Second, and related, any estimation strategy that exploits (1) and (OPT) will include (potentially testable) cross-equation restrictions. In that case, the demand parameters Λ are determined as a function of the parameters of the production function and adoption costs, M and N.

Now consider a condition under which we can speak unambiguously about the effect of increasing any of the practice-specific exogenous variables,

π is supermodular in $\left(y_j, X_j^1\right), \left(y_j, -X_j^0\right), \left(y_j, W_j^1\right)$, and $\left(y_j, -W_j^0\right)$ for all j and all Z, U. (ORD)

Up to a sign normalization, the restrictions on X in (ORD) are implicit in our definition of the model (1), and (ORD) simply extends these restrictions to the portion of profits not included in f. Under (ORD) and results from Milgrom and Roberts (1990), we can state conditions under which robust comparative statics predictions are available:[11]

Proposition 1 *Assume OPT, ORD. For fixed* Z *and* U, *if* π *is supermodular in* y, *then* y^* *is monotone nondecreasing in* $\left(W_j^1, X_j^1\right)$ *and monotone nonincreasing in* $\left(W_j^0, X_j^0\right)$.

The proposition states that if all of the choice variables are mutually complementary, an increase in the exogenous returns to one choice will lead to mutually reinforcing increases in all of the endogenously determined practices. This proposition will play an important role in our analyses of alternative econometric approaches in Sect. "Unobserved heterogeneity and reduced-form tests for complementarity".

Observability assumptions: the random systems model & the random practice model

Exogenous variation in the model arises from differences in the environment faced by each firm, such as the costs of factors of production, as well as features of the firm's location, market, and regulatory environment. If we wish to incorporate the possibility that firms choose practices to maximize profits in an application where the observables do not fully account for variation in practice adoption, it is necessary to accommodate the possibility that there are some exogenous costs and benefits to adoption which are observed by each firm but unobserved by the econometrician. This unobserved heterogeneity can be incorporated into the model by parsing each of the elements of exogenous variation $\left(\mathrm{X}^t, \mathrm{Z}^t, \mathrm{W}^t, \mathrm{U}^t\right)$ into an observed component (indicated in lowercase Roman typeface) and an unobserved component (lowercase Greek typeface). That is, we let $X_j^t = \left(\mathrm{x}_j^t, \boldsymbol{\chi}_j^t\right), Z_{\mathrm{k}}^t = \left(\mathrm{z}_{\mathrm{k}}^t, \boldsymbol{\zeta}_{\mathrm{k}}^t\right), W_j^t = \left(\mathrm{w}_j^t, \boldsymbol{\omega}_j^t\right)$, and $U_{\mathrm{k}}^t = \left(\mathrm{u}_{\mathrm{k}}^t, \boldsymbol{\nu}_{\mathrm{k}}^t\right)$. Note that measurement error in f can be incorporated by simply adding the same error to the unobserved return for all practices or systems.

Our interpretation of the data will depend critically on our assumptions and findings concerning the nature of unobserved heterogeneity $(\boldsymbol{\zeta}, \boldsymbol{\nu}, \boldsymbol{\chi}, \boldsymbol{\omega})$. We distinguish between two different models of the overall source of heterogeneity: models with incorporate system-specific shocks, $(\boldsymbol{\zeta}, \boldsymbol{\nu})$ (which we call the Random Systems Model or RSM); and models which accommodate only the practice-specific shocks $(\boldsymbol{\omega}, \boldsymbol{\chi})$ (a Random Practice Model or RPM). The RPM is simply a restriction on the RSM; in the two-choice case described in (1) and (1'), it requires that for a given $\boldsymbol{\zeta}$, there exists a $\boldsymbol{\chi}$ such that $\zeta_{00}^t + \zeta_{11}^t = \zeta_{01}^t + \zeta_{10}^t = \chi_1^{0t} + \chi_2^{0t} + \chi_1^{1t} + \chi_2^{1t}$, and likewise for $\boldsymbol{\nu}$ and $\boldsymbol{\omega}$. In other words, the unobserved return to any system can be composed into two parts, one for each practice, where these returns do not depend on choices about other practices.

Substantively, the main difference between these models is that the RPM imposes fixed coefficients on all practice

[9] Since elements of X^t and Z^t may be unobserved, the relationship between demand shocks and productivity will have important consequences for our empirical approach. McElroy (1986) examines the properties of this relationship for the standard production function with continuous capital and labor inputs, and shows that the errors in the factor demands have the interpretation as shocks to the marginal returns to factor inputs.

[10] This caveat is particularly important when f' is unobserved. In that case, inferences about complementarity will be based solely on revealed preference, and so assumptions about the adoption process will be critical.

[11] For additional discussion on the modeling issues associated with organizational design and complementarity, see, among others, Topkis (1978), Milgrom and Roberts (1990), and Athey et al. (1998).

interaction effects (unknown to the econometrician but observed by firms).[12] RPM might be an appropriate restriction when the differences across firms arise from regulatory or technological constraints associated with the exploitation of individual practices (such as work rule restrictions or the quality of equipment and facilities), while the RSM might be more appropriate when managers vary in their talent at exploiting combinations of practices. The RSM is required in the presence of an unobserved endogenous variable that interacts with more than one other choice, since its adoption will simultaneously change the value of adopting all other elements.

Although the RPM may seem like a reasonable assumption, it should be emphasized that specifying the organizational design production function entails ambiguities which, while present in the traditional problem, are more difficult to ignore in the context of organizational design. Most importantly, it is difficult to identify from theory what the "factors" of production are, and more difficult to argue that all factors have been appropriately accounted for. Thus, below we establish conditions under which the restrictions of the RPM can be tested.

Unobserved heterogeneity and reduced-form tests for complementarity

This section uses the formal model to examine the nature of the bias associated with several econometric procedures from the existing literature on complementarity in organizational design.[13] Our analysis highlights the precise assumptions that are required in order for these procedures to provide a consistent test statistic for complementarity. Section "Unobserved heterogeneity and econometric approaches to complementarity" reviews the two most common approaches used in the prior literature and shows that both are subject to bias due to unobserved heterogeneity. It also examines the extent to which different techniques exploiting instrumental variables alleviate these biases. In Section "Revealed preference reduced-form procedures", we argue that an approach based on revealed preference may be informative in cases in which specific pairwise interactions are particularly important and in which choice-specific instrumental variables are observed.

Unobserved heterogeneity and econometric approaches to complementarity

The first approach we examine is based on revealed preference (and thus implicitly makes an assumption of optimizing behavior). If y_i and y_j are complements, then, under the conditions of Proposition 1, a change in ω_i or χ_i will have a direct effect on the probability that y_i is adopted, which will in turn increase the probability that y_j is adopted. Thus, complementarity creates a force in favor of positive correlation (or "clustering") between y_i and y_j, even after controlling for observable, exogenous characteristics. This insight, analyzed theoretically in Holmstrom and Milgrom (1994), Arora and Gambardella (1990), and Arora (1996) motivates the use of the (CORR) approach:

> **(CORR)** Test whether the correlation among practices is positive, conditional on observables.

A substantial benefit to this approach is that it can be used even if only the adoption decisions are observed. Most of the recent empirical papers about complementarity use this approach, at least as supporting evidence.[14]

The second principal approach has been to build on the empirical productivity and firm performance literature. This approach relies on an OLS regression of a measure of performance/productivity on a set of regressors, including interaction effects between practices:

[12] The RPM is analogous to a standard assumption in the economics literature on productivity. Productivity models typically exploit duality, positing a "flexible" functional form for a firm's cost function which is linear in parameters but nonlinear in factor prices (Christensen et al. 1973; McElroy 1986; Jorgenson 1995). In such models, it is typical to assume that the elasticity of substitution between factors (capital and labor) is fixed across firms. The RPM assumption is analogous: the RSM allows the elasticity of substitution between practices to vary across firms, while the RPM holds the interaction effect fixed. The variables X_j and W_j can be interpreted as analogous to the "prices" for practice *j*. In addition to the discreteness of our model, it is the fact that some of these prices are unobserved which restricts us from using the duality approach.

[13] While our analysis will focus on procedures adopted from the traditional econometrics literature, other social scientists (most notably sociologists and psychologists) have attempted to measure the effects of organizational design using alternative statistical approaches. The principal alternative statistical procedures used by other social scientists can be implemented with the software package, LISREL (Joreskog and Sorbom 1995), which is used to estimate the parameters associated with systems of linear simultaneous equations, where firm practices may be discrete and measured with several indicator variables. LISREL does not explicitly account for the presence of unobserved heterogeneity and so is subject to the same biases associated with the procedures reviewed here.

[14] Arora and Gambardella (1990) develop (CORR) as a formal test for complementarity. Brickley (1999) explicitly uses (CORR) as a test for the comparative static predictions of Holmstrom and Milgrom (1994) in the context of franchising contract provisions. Other recent studies which use this approach, among others, include Ichniowski et al. (1997), Brynjolfsson and Hitt (1998), Colombo and Mosconi (1995), Greenan and Guellec (1998), Helper (1999), Helper and Levine (1994), Hwang and Weil (1996), Kelley et al. (1995), Macduffie (1995), Pil and Macduffie (1996), Leoni et al. (2001), Cassiman and Veugelers (2006), Cockburn et al. (2003).

(PROD) Estimate complementarity parameter from interaction effects in OLS estimation of the organizational design production function.

In the simple case of two practices, PROD requires the estimation of

$$f^t = \mathbf{1}^t_{00}\theta_{00} + \mathbf{1}^t_{01}\theta_{01} + \mathbf{1}^t_{10}\theta_{10} + \mathbf{1}^t_{11}\theta_{11} + \beta_1 x^t_1 y_{1t} + \beta_2 x^t_2 y^t_1 + \xi^t \quad (5)$$

where 1^t_{ij} is an indicator variable for observing the practice combination (i, j), and the corresponding test statistic for complementarity is $\kappa \equiv \theta_{11} - \theta_{01} - [\theta_{10} - \theta_{00}]$. Throughout the remainder of our analysis, we let $\hat{\kappa}^{oLS}$ represent the estimates of κ derived from OLS in (5). A version of PROD has been implemented by several authors, most notably in Ichniowski et al. (1997) (hereafter ISP), which presents a careful study of the impact on productivity of adopting different combinations of human resource practices in steel finishing lines. Like many such studies, ISP's data is gathered through detailed primary source surveys.[15]

A variant on (PROD) uses instrumental variables:

(PROD-IV) Estimate complementarity parameter from interaction effects in the organizational design production function using instrumental variables (IV).

This procedure requires choice-specific instrumental variables, and implementation requires an exclusion restriction, satisfied under the following assumption:

There exists a vector w such that, for all i, $\pi(\mathrm{y}, \mathrm{w}, \cdot)|_{y_i=1} - \pi(\mathrm{y}, \mathrm{w}, \cdot)|_{y_i=0}$ varies with w_i but not with w_j, for all $j \neq i$. (EXCL)

When implementing (PROD-IV), it will be necessary to construct an instrument for each system $\mathrm{y} \in \{0,1\}^J$. This can be accomplished by interacting the practice-specific instruments. As we discuss below, EXCL is not sufficient for the consistency of a reduced-form instrumental variables estimate of complementarity.

We discuss (CORR), (PROD), and (PROD-IV) with reference to different maintained hypotheses about the nature of unobserved heterogeneity and complementarity. One of the principal distinctions we will make is between cases where the practice-specific unobserved heterogeneity is composed of independent elements, versus affiliated elements. Affiliation is a strong form of correlation: a vector x of random variables is affiliated if it has a joint density, $g(\mathrm{x})$, such that $\log(g(\mathrm{x}))$ is supermodular. Affiliation implies that for all $i \neq j$ and all nondecreasing functions g_i and $g_j, \mathrm{cov}\left(g_i\left(x_i\right), g_j\left(x_j\right)\right) \geq 0$ (the latter property is called association), and further, this property holds conditional on any set of the form $\times_i [a_i, b_i]$.

For simplicity, in Sect. "Unobserved heterogeneity and reduced-form tests for complementarity" we will analyze the specific case in which the unobserved returns to choosing $y_j = 0$ are identically 0 , so that $\chi^{1,t}_j$ and $\omega^{1,t}_j$ are the unobserved returns and costs to adopting practice j.[16] In addition, we focus on a two-choice model, and we abstract away from system-specific variation, Z^t and U^t.[17] We define our main cases in Table 2.

The goal of the formal analysis in this section is to derive sufficient conditions under which the biases in the procedures (CORR), (PROD), and (PROD-IV) can be signed and interpreted. We begin by discussing (CORR) and (PROD), and return to discuss (PROD-IV).

Procedures not requiring instrumental variables

One of our main results is to identify plausible economic and/or managerial assumptions under which productivity (PROD) and adoption (CORR) deliver biases in the same direction. For example, when the unobserved returns to practice adoption are affiliated, then the conditional correlation between the practices will be positive, and further,

[15] ISP also mirror the more general literature in that their data is subject to extensive "clustering" so that many practice combinations are simply not observed (which preclude a direct estimate of the relevant parameters in (5)); their solution is to construct system "indexes" which aggregate over observed practice combinations.

Table 2 Assumptions about complementarity and practice-specific unobservables

Label	Description
Assumptions about complementarity	
TI	$\kappa = 0$
TC	$\kappa > 0$
Assumptions about practice-specific unobservables	
WI	$\omega^{1,t}$ is present, but $\omega^{1,t}$ is an independent vector and is independent of $\chi^{1,t}$.
WC	$\omega^{1,t}$ is present and strictly affiliated.
X0	$\chi^{1,t}$ is not present.
XI	$\chi^{1,t}$ is present, but $\chi^{1,t}$ is an independent vector.
XC	$\chi^{1,t}$ is present and strictly affiliated.

[16] We focus on this special case for expositional simplicity; Sect. "A structural approach to estimation and testing in organizational design" considers the general model where unobserved returns impact the firm whether or not each practice is adopted or not.

[17] The richness of the RSM is not required to establish the biases associated with (PROD) and (CORR), and the simple RPM allows us to evaluate the most interesting economic, managerial and statistical environments.

the estimates of interaction effects from OLS will also be biased upward. Thus, even if two distinct approaches are used to provide evidence about complementarity, one based on revealed preference (CORR) and the other on productivity (PROD), it is impossible to rule out alternative explanations. These findings further motivate the structural estimator developed in Sect. "A structural approach to estimation and testing in organizational design".

All of the propositions in this section will rely on the assumption that practices are optimally chosen (OPT). Without an assumption about practice adoption, it will be impossible to relate the primitives of the formal model to observable choices, a prerequisite to signing the biases from different procedures.[18] In other words, these propositions should be interpreted as suggestive of the biases that could potentially arise if we allow firms to respond to their environment in a way unobserved by the econometrician. It should be emphasized that despite its role in this section, the estimation procedure we propose in Sect. "A structural approach to estimation and testing in organizational design" does not require (OPT), and in Sect. "Testing theories about the adoption of organizational design practices" we discuss conditions under which (OPT) can be tested.

Finally, in order to compare the approaches (PROD) and (CORR), it will be useful to make the following assumption, requiring that there are no interactions between the practices outside the production function:

$g(\mathrm{y}, \mathrm{W}, \mathrm{U}; \mathrm{N}) = \pi(\cdot) - f(\cdot)$ is additively separable in y.

This assumption allows us to refer to a single measure of complementarity in our comparison of the CORR and PROD approaches.

When the unobserved exogenous variables take a particularly simple form, both PROD and CORR do indeed provide a consistent test for complementarity. Consider the case where (X0) and (WI) hold. In this restriction on the RPM, there is an idiosyncratic shock to the cost of adopting each practice, which is excluded from observed productivity, independent across practices, and unobserved by the econometrician.

Proposition 2 *Assume OPT, ORD, EXCL, NI, and* $\mathrm{Z} = \mathrm{U} = 0$. *Assume further X0 and WI* $\left(\chi^{1,t} = 0\right.$, *and* $\left.\boldsymbol{\omega}^{1,t}\right.$ *is an independent random vector). Then:*

(PROD) $E\left(\hat{\kappa}^{OLS}\right) = \kappa$.

(CORR) $\mathrm{Corr}\left(y_1, y_2 \mid \mathrm{x}, \mathrm{w}\right) \geq 0$ *if and only if TC is satisfied* $(\kappa \geq 0)$.

This proposition is intended to highlight the implicit assumptions underlying these procedures, and serve as a point of comparison for subsequent propositions. It is useful to consider scenarios under which these assumptions might be satisfied. Assumption (WI) corresponds to a scenario where there are random shocks to adoption or implementation costs that are unrelated across practices. Under X0, all of the unobserved heterogeneity enters outside the production function. This might occur if the production technology is similar across firms, the performance measure is a narrow one, and differences in adoption are driven by variation in the fixed cost of adopting practices. For example, ISP use a measure of assembly line productivity, and they provide qualitative evidence to support the hypothesis that their measure of performance is sufficiently narrow and their data detailed enough that they have ruled out many variables which fit our definition of χ*; the assumption they require is that the observed variation in practices in their data is the consequence of factors which do not directly affect their performance measure (i.e.,* w *and* $\boldsymbol{\omega}$*). It is useful to note that such an assumption will be harder to satisfy in applications in strategy or corporate finance where the performance measure is based on overall financial performance or long-term strategic performance.*

The next two propositions outline our results about the direction of biases when we introduce $\chi^{1,t}$. We begin with a formal analysis of this bias. Assuming a two-choice model and the simple structure for $\chi^{1,t}$ assumed above, the expected value of f^t conditional on y is as follows:

$$\begin{aligned} E\left[f^t \mid \mathrm{y}\right] = 1^t_{00}\theta_{00} + 1^t_{01} \cdot \left[\theta_{01} + E\left[\chi_2^{1,t} \mid \mathrm{y}^* = (0,1)\right]\right] \\ + 1^t_{10} \cdot \left[\theta_{10} + E\left[\chi_1^{1,t} \mid \mathrm{y}^* = (1,0)\right]\right] \\ + 1^t_{11} \cdot \left[\theta_{11} + E\left[\chi_2^{1,t} \mid \mathrm{y}^* = (1,1)\right]\right. \\ \left. + E\left[\chi_1^{1,t} \mid \mathrm{y}^* = (1,1)\right]\right] \end{aligned} \tag{6}$$

Because each element of $\chi^{1,t}$ only contributes to productivity (f) when the corresponding element of y^t is 1, $\hat{\theta}^{OLS}_{00}$ is unbiased, while $\hat{\theta}^{OLS}_{11}$ has two bias terms (one from each element of $\chi^{1,t}$). Under (OPT), these bias terms are nonzero. Figures 1 and 2 illustrate the regions of the unobservables that correspond to the optimal choices of each practice; notice that the shapes of these regions, and thus the biases, depend on whether or not the practices are complements.

In our application, our focus is not on any specific element of θ, but on the bias associated with κ, the complementarity coefficient. Thus, we are able to calculate the expected value of $\hat{\kappa}^{OLS}$ as follows:

[18] While it is certainly plausible that practice adoption is not affected by factors that impact the marginal returns to adoption but are unobserved by the econometrician, it should be recognized that precluding the hypothesis that choices respond to the economic environment is an assumption.

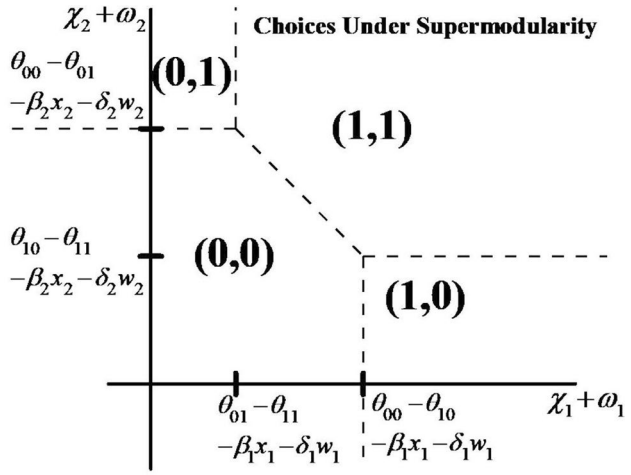


Fig. 1 Regions of optimal practice adoption under supermodularity

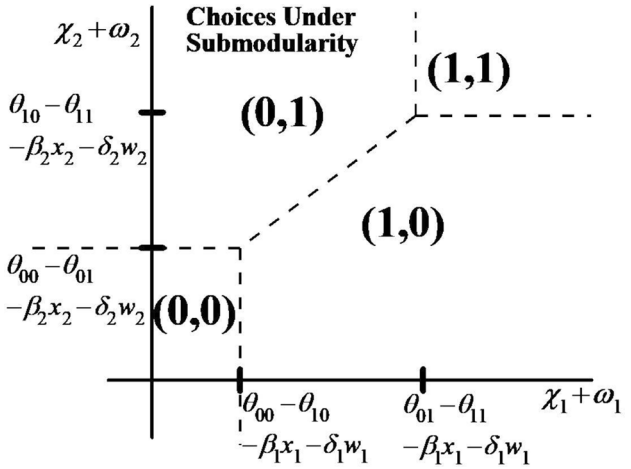


Fig. 2 Regions of optimal practice adoption under submodularity

$$
\begin{aligned}
E\left[\hat{\kappa}^{OLS}\right] = \kappa &+ E\left[\chi_1^{1,t} | \mathbf{y}^* = (1,1)\right] \\
&- E\left[\chi_1^{1,t} | \mathbf{y}^* = (1,0)\right] \\
&+ E\left[\chi_2^{1,t} | \mathbf{y}^* = (1,1)\right] \\
&- E\left[\chi_2^{1,t} | \mathbf{y}^* = (0,1)\right]
\end{aligned} \tag{7}
$$

The last four terms on the right-hand side of (7) will sum to zero only in the case when f is additively separable (i.e., TI) and the components of $\boldsymbol{\chi}$ are either uncorrelated (XI) or not present (X0).

Under several alternative economic assumptions, the bias in (7) can be signed. To begin, consider the assumptions (XI) and (WI), which require that the components of $\chi^{1,t}$ and $\boldsymbol{\omega}^{1,t}$ are independent.

Proposition 3 *Assume OPT, ORD, EXCL, NI, and* $\mathrm{Z} = \mathrm{U} = 0$. *Assume further WI and XI* $\left(\left(\omega^{1,t}, \boldsymbol{\chi}^{1,t}\right)\right.$ *is an independent random vector), and TC* $(\kappa > 0)$. *Then:*

(PROD) $E\left(\hat{\kappa}^{OLS}\right) < \kappa$.
(CORR) $\mathrm{corr}\left(y_1^t, y_2^t \mid \mathrm{x}^t, \mathrm{w}^t\right) \geq 0$.

Note that the degree of bias in the productivity equation is not bounded below by zero; even if $\kappa > 0$, it might easily turn out that $\mathrm{E}\left(\hat{\kappa}^{OLS}\right)$ is less than zero, yielding a test statistic which would (in expectation) reject complementarity when TC in fact holds. The intuition for the bias in (PROD) is that under XI and TC, when $y_2 = 1$, the returns to y_1 are higher. But in that case, a lower value of χ_1 will suffice to generate a choice of $y_1 = 1$ (see Fig. 1). Thus, the bias from OLS is negative. This is a generalization of the standard single-choice selection bias. The procedure (CORR) provides a consistent test of complementarity under WI and XI, since the only remaining force leading to positive correlation is that the practices are more productive when adopted together.

Alternatively, consider a case where TI is satisfied (practices are independent from one another in f) but the unobserved productivity gains from each practice are affiliated (XC). Such a form for unobserved heterogeneity is plausible in many economic environments. For example, it might arise when the benefits associated with adopting each practice depends on the skill of a single manager or group of workers. In this case, OLS as well as CORR will overestimate the complementarity parameter.

Proposition 4 *Assume OPT, ORD, EXCL, NI, and* $\mathrm{Z} = \mathrm{U} = 0$. *Assume further W0*[19] *and XC* $\left(\boldsymbol{\omega}^{l,\mathrm{t}} = 0\right.$, *and* $\chi^{1,\mathrm{t}}$ *is strictly affiliated), and TI* $(\kappa = 0)$. *Then:*

(PROD) $E\left(\hat{\kappa}^{OLS}\right) > \kappa = 0$.
(CORR) $\mathrm{corr}\left(y_1^t, y_2^t \mid \mathrm{x}^t, \mathrm{w}^t\right) > 0$.

Proposition 4 highlights a fundamental identification problem for testing complementarity: both CORR and PROD find complementarity (erroneously) under the same assumptions about the economic and managerial environment. Many features of an organization could induce affiliation between the costs and benefits of practices, including the corporate culture and the characteristics of the workforce. The intuition behind Proposition 4 is straightforward: practice 1 is adopted when its unobserved returns are high. But this tends to happen exactly when the unobserved returns to practice 2 are also high, since the unobserved returns are positively correlated, and thus the expected value of the

[19] This result and Proposition 5 can be generalized to allow for WC rather than W0, but some additional regularity conditions are required.

unobserved returns to practice 2 are higher when practice 1 is adopted as well.

It is useful to note that several existing studies attempt to document complementarity by providing (independent) evidence in the spirit of CORR and PROD.[20] *Together, Propositions 2–4 show that any interpretation of the OLS estimates require a maintained assumption about the unobservables. Under more general structures, the sign of the correlation between the unobserved exogenous variables can either lead to a finding that complementarities don't exist when they really do, or a finding of complementarity when the choices are technologically independent. Using similar logic, Propositions 3 and 4 imply that a finding that* $\kappa < 0$ *but* $\text{corr}\left(y_1^t, y_2^t \mid \mathrm{x}^t, \mathrm{w}^t\right) \geq 0$ *also has an ambiguous interpretation: it is consistent with the hypothesis that the choices are substitutes, but* $\chi^{1,t}$ *is affiliated, as well as the hypothesis that the choices are complements, and* $\chi^{1,t}$ *is independent.*

Finally, our analysis has implications for the interpretations of the bias associated with analyzing the effects of a single choice on performance when a second endogenous practice is unaccounted for by the econometrician. For example, if we wish to study the effects of training programs on performance, the interpretations of a regression of performance on the adoption of training depends on whether other practices (such as the adoption of IT) contribute to productivity, are complementary with training, and are subject to unobserved heterogeneity in adoption. In our two-choice example, suppose that the econometrician observes all of the relevant features which affect the returns to y_1 *(so that* $\chi_1^{1,t} = 0$*) and that we are interested in measuring* $\theta_{11} - \theta_{01}$*. Suppose further that we observe* y_2*.*[21] *The presence of unobserved returns to* y_2 *will lead to an OLS estimate of the returns to* y_1 *conditional on* $y_2 = 1$*, as follows:*

$$\begin{aligned}\hat{\theta}_{11}^{oLS} - \hat{\theta}_{01}^{oLS} &= E\left[f^t|\mathbf{y}^t = (1,1)\right] \\ &\quad - E\left[f^t|\mathbf{y}^t = (0,1)\right] \\ &= \theta_{11} - \theta_{01} + E\left[\chi_2^{1,t}|\mathbf{y}^t = (1,1)\right] \\ &\quad - E\left[\chi_2^{1,t}|\mathbf{y}^t = (0,1)\right]\end{aligned} \tag{8}$$

Even in such a simple case, the crucial parameter of interest will be biased in the face of unobserved heterogeneity:

Proposition 5 *Assume OPT, ORD, EXCL, NI, and* $\mathrm{Z} = \mathrm{U} = 0$*.*

(a) *Assume further W0 and* XI $\left(\boldsymbol{\omega}^{1,\mathrm{t}} = 0\right.$*, and* $\left.\boldsymbol{\chi}^{1,\mathrm{t}}\right.$ *is an independent random vector), and TC (*$\kappa > 0$*). Then* $\hat{\theta}_{11}^{OLS} - \hat{\theta}_{01}^{OLS} < \theta_{11} - \theta_{01}$*.*

(b) *Assume further W0 and* XC $\left(\boldsymbol{\omega}^{l,\mathrm{t}} = 0\right.$*, and* $\left.\boldsymbol{\chi}^{1,\mathrm{t}}\right.$ *is strictly affiliated), and* $TI(\kappa = 0)$*. Then* $\hat{\theta}_{11}^{OLS} - \hat{\theta}_{01}^{OLS} > \theta_{11} - \theta_{01}$*.*

Proposition 5 highlights why understanding the organizational design production function is important, even when only one practice is of immediate policy relevance. In particular, without considering the full set of organizational practices, one will routinely infer a biased estimate of the returns to individual practices.

Instrumental variables procedures

We now consider whether instrumental variables procedures can be used to overcome the biases identified in the previous subsection. In other words, if the endogeneity of y_1 and y_2 are a source of bias, do standard instrumental variables procedures yield consistent tests for complementarity, and if so, under what conditions? For the types of unobserved heterogeneity analyzed in the previous section (e.g., XC), the simple answer to this question is no. Essentially, the traditional argument for the use of 2SLS does not apply here, unless χ is degenerate (X0). More precisely, the standard conditions for consistency of 2SLS are violated even if (EXCL) is satisfied. Since the unexplained portion of the productivity equation is given by $\chi_1^{1,t} y_1^t + \chi_2^{1,t} y_2^t$, and since y is endogenously determined as a function of all of the exogenous variables, (x, w, χ, and ω), w is not uncorrelated with the error term. Because unobserved heterogeneity in organizational design choice problems impacts both the adoption decision and realized performance, a 2SLS productivity regression equation will still be subject to bias.

To gain some intuition for the sources and directions of these biases, we examine an instrumental variables approach in which we can calculate a consistent test statistic for complementarity, under the null hypothesis of no interaction effects (TI). For each practice, assume that the econometrician observes a single binary practice-specific instrumental variable (i.e., each element of w is a dummy variable). Expanding on the logic of Figs. 1 and 2, we illustrate the case of no complementarity $(\kappa = 0)$ and binary instrumental variables in Fig. 3, where the Figure illustrates the optimal choices of y_1 and y_2 for different regions of unobservables $\left(\boldsymbol{\chi}^{1,t}\right.$ and $\left.\boldsymbol{\omega}^{1,t}\right)$ given w_1 and w_2.

Under the null hypothesis of no interaction effects (TI), a calculation based on Local Average Treatment Effects (LATE, as defined in Imbens and Angrist (1994)) can yield a consistent test for complementarity. The logic is somewhat

[20] ISP and Brynjolfsson and Hitt (1998) are two examples.

[21] The analysis is similar if y_2 is unobserved, but we would need to consider the average returns to y_1.

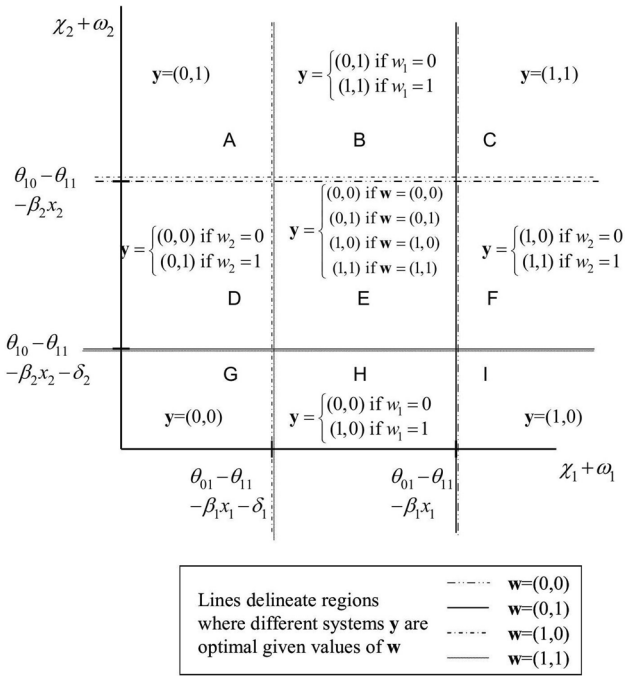


Fig. 3 Regions of adoption for different values of the instrumental variables

subtle, and we argue below that it does not extend to the case where choices do interact in determining productivity. To construct our test statistic, we derive an estimate of the effect of y_1 on productivity, using w_1 as an instrument for y_1 for the subset of firms where $w_2 = 0$ (without controlling for y_2). This estimate is just the LATE of y_1 on f for the subsample of the data for which $w_2 = 0$; as derived in Imbens and Angrist (1994), this is equal in expectation to :

$$\psi_0 \equiv \frac{E\left[f^t \mid w_1^t = 1, w_2^t = 0\right] - E\left[f^t \mid w_1^t = 0, w_2^t = 0\right]}{\Pr\left(y_1^t = 1 \mid w_1^t = 1, w_2^t = 0\right) - \Pr\left(y_1^t = 1 \mid w_1^t = 0, w_2^t = 0\right)}$$

The LATE parameter ψ_0 is the average effect of practice y_1 on the firms that were induced to adopt practice y_1 as a result of increasing w_1 from 0 to 1 (conditional on $w_2 = 0$). In Fig. 3, this corresponds to the change in performance associated with firms with χ and ω in region B switching from $\mathrm{y} = (0, 1)$ to $\mathrm{y} = (1, 1)$ and firms in regions E and H switching from (0, 0) to (1, 0). An important thing to notice is that under (TI), w_1 does not affect firm choices about y_2. Then, we can rewrite the LATE formula in this specific case as:

$$\psi_0 = \left(\theta_{11} - \theta_{01} + E\left[\chi_1 \mid B\right]\right) \frac{\Pr(B)}{\Pr(B) + \Pr(E) + \Pr(H)} + \left(\theta_{10} - \theta_{00} + E\left[\chi_1 \mid E \cup H\right]\right) \frac{\Pr(E) + \Pr(H)}{\Pr(B) + \Pr(E) + \Pr(H)}$$

ψ_0 is simply the (appropriately weighted) average return to y_1 in regions B, E, and H.

Now consider repeating the above procedure for the subsample of the data where $w_2 = 1$. Once again, we can use 2SLS to estimate the LATE of y_1 on productivity (using w_1 as an instrument) for those firms where $w_2 = 1$, which is given by:

$$\psi_1 \equiv \frac{E\left[f^t \mid w_1^t = 1, w_2^t = 1\right] - E\left[f^t \mid w_1^t = 0, w_2^t = 1\right]}{\Pr\left(y_1^t = 1 \mid w_1^t = 1, w_2^t = 1\right) - \Pr\left(y_1^t = 1 \mid w_1^t = 0, w_2^t = 1\right)}$$

This expression is similar to the one for ψ_0, except that different subsets of firms are induced to adopt. This expression can be written as

$$\psi_1 = \left(\theta_{11} - \theta_{01} + E\left[\chi_1^{1,t} \mid B \cup E\right]\right) \frac{\Pr(B) + \Pr(E)}{\Pr(B) + \Pr(E) + \Pr(H)} + \left(\theta_{10} - \theta_{00} + E\left[\chi_1^{1,t} \mid H\right]\right) \frac{\Pr(H)}{\Pr(B) + \Pr(E) + \Pr(H)}$$

A simple application of Bayes' rule implies that

$$\psi_1 - \psi_0 = \frac{\Pr(E)}{\Pr(B) + \Pr(E) + \Pr(H)} \left(\theta_{11} - \theta_{01} - (\theta_{10} - \theta_{00})\right) = 0$$

Intuitively, the difference between the two LATE estimators gives the average complementarity parameter for firms in region E, the group affected differently by w_1 when $w_2 = 1$ versus $w_2 = 0$. However, because (TI) implies that $\theta_{11} - \theta_{01} = \theta_{10} - \theta_{00}$, increasing the benefit to y_2 does not affect the returns to increasing the benefit of y_1, and the comparison of LATE estimators captures that.[22] This discussion motivates the following proposition:

Proposition 6 *Assume OPT, ORD, and EXCL. Then, under TI, the difference in the LATE parameters is equal to* $0 : \psi_1 - \psi_0 = 0$.

At first glance, Proposition 6 seems encouraging for the use of reduced-form instrumental variables procedures. However, on closer examination, the case considered in Fig. 3 is quite special; in all but a small number of cases, instrumental variables approaches will yield a biased test statistic. While Proposition 6 holds under the maintained hypothesis of no interaction effects (TI), relaxing this assumption results in a complex set of biases (Appendix B includes a more complete analysis). In particular, $\psi_1 - \psi_0$ can be positive, negative, or zero in the presence of

[22] This is true even if the unobserved returns are correlated, For example, if the unobserved returns are positively correlated (XC and WC), the average value of $\chi_1 + \omega_1$ would be higher in region B than in E, and higher in E than in H. However, because we are assuming TI, the expectation of the proposed test statistic would still be equal to zero, and so the distribution of the unobservables does not directly impact the analysis.

complementarity (TC). To see why, recall that in the case of (TI), w_1 does not affect choices about y_2. Under (TC), w_1 affects choices about both y_1 and y_2. As described in Appendix B, multiple selection effects impact the estimation, and the sign of the bias depends on the degree of complementarity and the joint distribution of the unobserved returns. In addition, the difference between the LATE estimators cannot be interpreted as the average degree of complementarity for any particular subpopulation; instead, it is the weighted average of differences between the average returns to individual practices for distinct subpopulations of firms.

In sum, although a rejection of the hypothesis that $\psi_1 - \psi_0 = 0$ is evidence that $\kappa \neq 0$, the sign of $\psi_1 - \psi_0$ is not necessarily informative about whether practices are substitutes or complements. In addition, we may fail to reject $\psi_1 - \psi_0 = 0$ when practices are in fact substitutes or complements, if the selection biases balance the impact of complementarity.

Finally, it is important to highlight that $\psi_1 - \psi_0$ is not the parameter estimated using (PRODIV), where we use w_1, w_2, and $w_1 \cdot w_2$ as instruments for y_1, y_2, and $y_1 \cdot y_2$ in a productivity regression. The expression for the coefficient on the interaction term $y_1 \cdot y_2$ using 2SLS is quite complex, depending on, among other things, the correlation structure of the unobservables and the correlation among the instruments. Despite this, our analysis of $\psi_1 - \psi_0$ provides some useful intuition about the types of biases we would expect in (PROD-IV).

Revealed preference reduced-form procedures

A final approach in the prior literature exploits revealed preference and the satisfaction of (EXCL) for some of the choice variables. Consider the reduced-form regression

$$y_i^t = \alpha_i^t + \lambda_i x_i^t + \delta_i w_i^t + \lambda_j x_j^t + \delta_j w_j^t + \varepsilon_j \tag{9}$$

As shown by Arora (1996), Proposition 2 implies the following:

Proposition 7 *Assume OPT, ORD,EXCL, NI, and* $\mathrm{Z} = \mathrm{U} = 0$. *Assume further* $TC(\kappa > 0)$. *Then* $E[y_i \mid \mathrm{x}, \mathrm{w}]$ *is nondecreasing in* w_j.

Further, under (EXCL), in a two-choice model, only complementarity leads to a positive effect of w_i on y_j: a factor which has its sole direct effect on y_j will be uncorrelated with y_i unless y_i and y_j interact directly in the production function. This motivates:

(ADP) Reduced-form tests of adoption exploiting exclusion restrictions (as in (9)).

This approach is exploited in several studies. For example, Slade (1996) presents evidence for complementarity in retail gasoline stations by examining how gasoline contract choice depends on the other features of the retailing environment, Brickley (1999) provides evidence about how several features of franchising contracts change with the degree to which the franchisee relies on repeat business, and Novak and Stern (2009) provide evidence that vertical integration choices within automobile product development systems are sensitive to exogenous measures associated with other (potentially complementary) systems within the same vehicle model.

When there are only two practice choices, and when choice-specific instrumental variables are only available for one of these practices, (ADP) does provide a simple and consistent test for complementarity. By taking advantage of (EXCL) in the adoption equation for y_j, (ADP) relies crucially on revealed preference and the exogeneity of w_i. Moreover, as discussed in Arora (1996), (ADP) is subject to qualification when there are more than two choice variables. Consider the case of three choice variables where the following relationships hold: (y_2, y_3) are complements, (y_1, y_3) are complements, and (y_1, y_2) are substitutes. Under these conditions, an increase in w_1 might lead to an increase in all three choices, if the effects through the chain $y_1 \to y_3 \to y_2$ outweigh the effects through the chain $y_1 \to y_2$. In this case, (ADP) cannot provide a precise test for pairwise complementarity. However, if the instrumental variables are valid, we can test whether all pairs are complements (i.e., this can be rejected if the reduced-form coefficients are sufficiently negative). In sum, (ADP) provides indirect evidence for complementarity, motivating the structural approach described in the next section.

A structural approach to estimation and testing in organizational design

Identification and the nature of unobserved heterogeneity

This section presents the main identification and testing theorems of the paper, providing assumptions under which the key parameters of the RSM and RPM models are identified, as well as assumptions under which it is possible to test for complementarity and test the assumptions of the RPM model (as nested within the RSM model). The most general result specifying the conditions for identification of the RSM model draws from a recent literature on nonparametric identification. Our contribution is to develop the precise linkage between this (abstract) econometric literature and specific problem of evaluating the impact of organizational design on performance. The results concerning identification

of parameters within the RPM model are more novel. We show that practice-specific instrumental variables are sufficient for identification, even though when practices are complementary, practice-specific instruments do not satisfy the exclusion restrictions that are typically required in the literature (e.g., Thompson (1989)). In particular, we derive the conditions for identification of the key parameters of this specific model, and link identification to the observability of choice-specific instrumental variables (EXCL).

While the theorems in this section are somewhat technical, these results have implications for researchers doing empirical work, particularly for those involved in data gathering or survey design. For example, the analysis demonstrates that the identification of complementarity assuming the RPM requires the observability of choice-specific instrumental variables (EXCL) as well as an appropriate correction for selectivity. In contrast, testing the assumptions of the RPM (which is nested within the more general RSM model) requires the observability of system-specific instrumental variables. For self-gathered datasets, researchers can trade off data gathering costs against the benefits of observing variation of one type or the other.

Finally, it is useful to emphasize that, although our analysis does not make functional form assumptions about the distribution over unobservables, the propositions cannot in general be sharpened by such assumptions (e.g., by assuming a multivariate normal distribution). By avoiding distributional assumptions in our identification results, we are able to clarify how variation in exogenous variables enables the econometrician to learn about the structural parameters. Although parametric models may often be more convenient for estimation, the researcher can have greater confidence about the results of a suitably flexible parametric model knowing that identification does not hinge on specific functional form assumptions.

Consider a general model of productivity with discrete choices is given as follows (where 1_{y}^{t} is an indicator system y).[23]

$$f\left(\mathrm{y}^{t};\mathrm{Z}^{t},\boldsymbol{\alpha}\right)\equiv\zeta_{\mathrm{y}^{t}}^{t}+\boldsymbol{\alpha}_{\mathrm{y}^{t}}\cdot\mathrm{z}^{t} \tag{10}$$

We let $\theta_{\mathrm{y}}\equiv E\left[\zeta_{\mathrm{y}}^{t}\right]$ so that θ_{y} can be used to construct tests about the average level of complementarity, net of z. A discrete choice model for adoption can be built around the following specification for the decision-maker's utility from each system:

$$\pi_{\mathrm{y}}^{t}=\eta_{\mathrm{y}}\mathrm{z}^{t}+\varphi_{\mathrm{y}}\mathrm{u}^{t}+\zeta_{\mathrm{y}}^{t}+\nu_{\mathrm{y}}^{t} \tag{11}$$

We let $G(\zeta+\nu)$ be the joint distribution of $\zeta+\nu$, and let $F(\zeta)$ be the joint distribution of ζ.

A number of authors have studied this problem; here, we state the identification results slightly differently from the literature so that we may tailor the results to our model. The main result in the literature is that if there is sufficient variation in exogenous variables that differentially affect the returns to different systems, the joint distribution of unobserved returns can be identified up to scale and location. To gain some intuition for this, let $\Delta\zeta_{\mathrm{y}}^{t}=\zeta_{\mathrm{y}}^{t}-\zeta_{0}^{t}$ and let $\Delta\nu_{\mathrm{y}}^{t}=\nu_{\mathrm{y}}^{t}-\nu_{0}^{t}$, where 0 represents the system (0, .., 0), and let G^{Δ} be the joint distribution of $\Delta\zeta_{\mathrm{y}}^{t}+\Delta\nu_{\mathrm{y}}^{t}$. Suppressing z for simplicity, if u is independent of $\Delta\zeta_{\mathrm{y}}^{t}+\Delta\nu_{\mathrm{y}}^{t}$ then

$$\begin{aligned}\Pr\left(\mathrm{y}^{t}=0\mid\mathrm{u}^{t}\right)&=\Pr\left(\pi_{0}^{t}\geq\pi_{\mathrm{y}'}^{t}\text{ for all }\mathrm{y}'\neq0\right)\\&=\Pr\left(\left(\varphi_{\mathrm{y}'}-\varphi_{0}\right)\mathrm{u}^{t}\geq\Delta\zeta_{\mathrm{y}'}^{t}+\Delta v_{\mathrm{y}'}^{t}\text{ for all }\mathrm{y}'\neq0\right)\\&=G^{\Delta}\left(\left(\varphi_{(0,\ldots1)}-\varphi_{0}\right)\mathrm{u}^{t},\ldots,\left(\varphi_{(1,\ldots,1)}-\varphi_{0}\right)\mathrm{u}^{t}\right)\end{aligned} \tag{12}$$

Clearly, multiplying each π_{y}^{t} by a positive constant does not change the probability that any system is observed. So, it is necessary to normalize the scale of the unobservables. Then, (12) illustrates that G^{Δ}, the distribution of $\Delta\zeta_{\mathrm{y}}^{t}+\Delta\nu_{\mathrm{y}}^{t}$, is identified (up to scale) at any point $\mathrm{c}\in\Re^{2^{J}-1}$ in its support if and only if there is some value of u such that

$$\left(\left(\varphi_{(0,\ldots,1)}-\varphi_{0}\right)\mathrm{u},\ldots,\left(\varphi_{(1,\ldots1)}-\varphi_{0}\right)\mathrm{u}\right)=\mathrm{c}$$

Define

$$\Delta\Phi\equiv\begin{bmatrix}\left(\varphi_{(0,\ldots,1)}-\varphi_{0}\right)\\\vdots\\\left(\varphi_{(1,\ldots,1)}-\varphi_{0}\right)\end{bmatrix}$$

Then, if $\Delta\zeta_{\mathrm{y}}^{t}+\Delta v_{\mathrm{y}}^{t}$ has support equal to $\Re^{2^{J}-1}$, identification of G^{Δ} will hold if, for example, each component of u varies over the real line, and $\mathrm{rank}(\Delta\Phi)\geq2^{J}-1$. More generally, G^{Δ} can be identified in regions that can be "traced out" through variation in u.

Formalizing this discussion, the following will be sufficient for identification of G^{Δ}.[24]

Assumption 1 Assume that (a) (z,u) is independent of (ζ,ν),[25] (b) $\Pr(\mathrm{u}\in E\mid\mathrm{z})>0$ for all $E\in\Re^{m}$,

[23] For concision, we suppress X (such variables can be incorporated in Z). As well, for simplicity we restrict attention to models that are linear and additive in the exogenous variables. Much of the analysis extends when we replace $\boldsymbol{\alpha}_{\mathrm{k}}\cdot\mathrm{z}^{t}$ with a nonlinear, unknown function of z^{t}; see, e.g., Matzkin (1990, 1992) or Das et al. (2003).

[24] Thompson (1989) provides similar conditions for identification; he states his conditions in terms of exclusion restrictions that may not be directly satisfied in our application. This motivates our alternative formulation.

[25] If z is not independent of (ζ,ν), then identification of (ζ^{t},ν^{t}) conditional on z can be established if the conditions of Assumption 1 hold conditional on z.

where $m \geq 2^J - 1$ is the dimensionality of u, and (b) $\text{rank}(\Delta\Phi) \geq 2^J - 1$.

An important observation is that Assumption 1 is not satisfied when the only observed instrumental variables are practice-specific (rather than system-specific).[26] While practice-specific instrumental variables can be rewritten as system-specific instrumental variables, a single practice-specific instrumental variable impacts multiple systems. Although Assumption 1 does not directly require any exclusion restrictions, the problem is that the impact of the set of practice-specific instruments on, say, system (1, 1) is a linear combination of its effect on (0, 0), (0, 1), and (1, 0). But, in order to trace out the joint distribution of returns to systems, we must observe variables that shift the returns to the whole set of systems in arbitrary ways (except perhaps the "base" or reference system).

The identification result for the RSM model, however, requires Assumption 1:

Proposition 8 *Maintain Assumption 1. Further:*

(a) *Assume (10), (11), and that choices are determined to maximize payoffs as given by (11). Suppose that* $\mathrm{y}^t, \mathrm{z}^t$, *and* u^t *are observed, while* (ζ^t, ν^t) *are unobserved.*
(b) *Let* $\mathcal{F}$ *and* $\mathcal{G}$ *be sets of absolutely continuous probability distributions, and suppose that* $G \in \mathcal{G}$, *and, for each* $\mathrm{y} \in \{0,1\}^J$, $F_{\mathrm{y}} \in \mathcal{F}$. *Let* $\mathrm{F} = (F_{0,\ldots,}, .., F_{1,.,1})$.
(c) *Suppose* $(\eta, \varphi) \in \mathscr{L}$ *and* $\boldsymbol{\alpha} \in \mathcal{M}$, *where* $\mathcal{L}$ *and* $\mathcal{M}$ *are compact.*

Then:

(i) *If f is observed, then* $(\boldsymbol{\alpha}, \eta, \varphi, \mathrm{F}, G)$ *are identified in* $\left(\mathfrak{L}, \mathcal{M}, \mathcal{F}^{2^J}, \mathcal{G}\right)$ *up to the scale of* (η, φ, ν^+) *and location of* ν^t.
(ii) *If f is unobserved, then* $(\boldsymbol{\eta}, \boldsymbol{\varphi}, G)$ *is identified in* $(\mathcal{L}, \mathrm{II})$ *up to the scale of* $(\boldsymbol{\eta}, \boldsymbol{\varphi}, \zeta^t + \boldsymbol{\nu}^t)$ *and the location of* $\zeta^t + \nu^t$.

[26] To see this, suppose that there are two practices, and suppose that $\varphi_{\mathrm{k}}\mathrm{u}^t = \delta_1^{k_1} w_1^{k_1,t} + \delta_2^{k_2} w_2^{k_2,t}$, where, for example, $\delta_1^0 w_1^{0,t}$ enters the firm's payoff function when practice 1 is not adopted, while $\delta_1^1 w_1^{1,t}$ enters the firm's payoff when practice 1 is adopted. Then,

$$\begin{aligned}\varphi_{11}\mathbf{u}^t =& \delta_1^1 w_1^{1,t} + \delta_2^1 w_2^{1,t} = \delta_1^0 w_1^{0,t} \\ &+ \delta_2^1 w_2^{1,t} + \delta_1^1 w_1^{1,t} + \delta_2^0 w_2^{0,t} \\ &- \left(\delta_1^0 w_1^{0,t} + \delta_2^0 w_2^{0,t}\right) \\ =& \varphi_{01}\mathbf{u}^t + \varphi_{10}\mathbf{u}^t - \varphi_{00}\mathbf{u}^t\end{aligned}$$

so that $\varphi_{11} - \varphi_{00} = (\varphi_{01} - \varphi_{00}) + (\varphi_{10} - \varphi_{00})$. violating the rank condition of Assumption 1 .

Proposition 8 establishes the conditions under which we can identify the mean of Z^t, which in turn implies that an estimate for "average complementarity" is available. Part (ii) was established in the text above, and part (i) follows directly from, e.g., Ichimura and Lee (1991) or Heckman et al. (1997).[27] A comparison of parts (i) and (ii) isolates the effect of observing productivity. When productivity is unobserved, we can test for the sign of the average complementarity between two practices, but the magnitude cannot be identified.

Under the assumptions of the RPM, Assumption 1 is not necessary for the identification of complementarity or the distribution of unobserved returns. We derive the identification of the parameters for the RPM by first imposing the restrictions on $G\left(\zeta^t + \nu^t\right)$ and $\mathrm{F}\left(\zeta^t\right)$ imposed by the RPM, as follows:

$$\zeta_{\mathrm{y}}^t + \nu_{\mathrm{y}}^t = \sum_{j=1}^{J} \chi_j^{y_j,t} + \sum_{j=1}^{J} \omega_j^{y_j,t} \text{ and } \zeta_{\mathrm{y}}^t = \sum_{j=1}^{J} \chi_j^{y_j,t} \tag{13}$$

Equation (13) places restrictions on the joint distribution $G\left(\zeta^t + \nu^t\right)$. Suppose that G is identified up to location and scale, so that G^{Δ} is identified up to scale. The following is therefore a consequence of (RPM):

$$\Delta\zeta_{\mathrm{y}}^t + \Delta\nu_{\mathrm{y}}^t = \sum_{j=1}^{J} 1_{y_j=1} \cdot \Delta\chi_j^t + \sum_{j=1}^{J} 1_{y_j=1} \cdot \Delta\omega_j^t \tag{14}$$

where the incremental returns to a practice j are denoted $\Delta\chi_j^t = \chi_j^{1,t} - \chi_j^{0,t}$ and $\Delta\omega_j^t = \omega_j^{1,t} - \omega_j^{0,t}$.

Our identification strategy relies on practice-specific instrumental variables that satisfy (EXCL). Formally:

Assumption 2 Assume that (a) the model is given by (10), (11), (13), and (14), and that firms choose practices to maximize the payoffs as given by (11); (b) (z, w) is independent of (χ, ω), (c) for all $\mathrm{z}, \Pr\left(\left(\delta_1^1 w_1^{1,t} - \delta_1^0 w_1^{0,t}, \ldots, \delta_J^1 w_J^{1,t} - \delta_J^0 w_J^{0,t}\right) \in E \mid \mathrm{z}\right) > 0$ for all $E \in \Re^J$.

Assumption 2 will be satisfied even if $w_j^{0,t} = 0$ for all $j = 1, .., J$, so long as the support of $\left(w_1^{1,t}, \ldots, w_J^{1,t}\right)$ is

[27] Note that only the marginal distributions of ζ are identified. Intuitively, the systems are never observed together, and so the covariance between the returns cannot be recovered. Heckman and Honore (1990) show that under the extreme assumption that $v \equiv 0$, the full joint distribution of $F(\zeta)$ is identified if f is observed under an assumption analogous to Assumption 1. In the current context, this is equivalent to assuming away unobserved shocks to the costs of adoption that are not incorporated into the productivity measure, f.

$\Re^J$ and $\delta_j^1 \neq 0$ for all $j = 1, .., J$. In words, Assumption 2 is satisfied if we have practice-specific instruments with full support (which rules out perfect correlation among the instruments, but allows some correlation among them), so long as the instruments have some non-zero impact on the returns to the relevant practices. Assumption 2 implies (EXCL).

To state our identification result, it is useful to introduce notation for the level of complementarity between two practices, conditional on the choices about other practices, as follows (where, for example, $\theta_{11,y_{-ij}}$ is the return when $y_i = 1$ and $y_j = 1$ given y_{-ij}):

$$\kappa_{ij}\left(\mathrm{y}_{-ij}\right) = \left(\theta_{11,\mathrm{y}_{-ij}} - \theta_{01,\mathrm{y}_{-ij}}\right) - \left(\theta_{10,\mathrm{y}_{-ij}} - \theta_{00,\mathrm{y}_{-ij}}\right)$$

Proposition 9 *Maintain Assumption 2, as well as (a), (b), and (c) of Proposition 8.*

(i) *If f is observed, then for all $i \neq j$, the complementarity between practices i and $j, \kappa_{ij}\left(\mathrm{y}_{-ij}\right)$, is identified for each y_{-ij}. The marginal distribution of $\zeta_{\mathrm{y}}^t = \sum_{j=1}^J \chi_j^{y_j,t}$ is identified up to location, and $\mathrm{cov}\left(\Delta\chi_i^t, \Delta\chi_j^t\right)$ is identified.*

(ii) *If f is unobserved, then for all $i \neq j$, the sign of $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$ is identified for each y_{-ij}. The joint distribution of $\left(\Delta\chi_i^t + \Delta\omega_i^t, \Delta\chi_j^t + \Delta\omega_j^t\right)$ (marginalized over $\Delta\chi_{-ij}^t$) is also identified up to location and scale.*

Proposition 9 states that the main objects of economic interest in the RPM are identified under Assumption 2. In particular, when productivity and choice-specific instrumental variables are observed, we can identify the level of complementarity among each pair of practices, as well as the level of correlation among unobserved returns to the practices. When productivity is unobserved, for each pair of practices, we can identify the sign of complementarity as well as whether unobserved returns are positively or negatively correlated.[28]

Finally, linking Propositions 8 and 9 together, it is possible under Assumption 1 to distinguish between the two different forms of unobserved heterogeneity. Testing the RPM may be important in many empirical settings since when the RPM holds, it is possible to draw unambiguous policy conclusions about interaction effects, and interpret the selection biases of OLS and CORR analyses according to the propositions in Sect. "Unobserved heterogeneity and reduced-form tests for complementarity". One possible test is based on a direct implication of (14):

$$\mathrm{var}\begin{pmatrix} \Delta\nu_{11,\mathbf{y}_{-ij}}^t + \Delta\nu_{00,\mathbf{y}_{-ij}}^t - \Delta\nu_{01,y_{-ij}}^t - \Delta\nu_{10,\mathbf{y}_{-ij}}^t \\ +\Delta\zeta_{11,\mathbf{y}_{-ij}}^t + \Delta\zeta_{00,\mathbf{y}_{-ij}}^t - \Delta\zeta_{01,\mathbf{y}_{-ij}}^t - \Delta\zeta_{10,\mathbf{y}_{-ij}}^t \end{pmatrix} = 0 \text{ for all } i, j, \mathbf{y}_{-ij} \tag{15}$$

This is a testable restriction under Proposition 8. Formally:

Proposition 10 *Suppose $G\left(\mathrm{Z}^t + \mathrm{u}^t\right)$ is known up to location and scale.*[29] *Then RPM is testable.*

We can use these identification results to develop a specific estimator to test for complementarity, which is therefore the next task to which we turn.

A structural estimator for the organizational design production function

This section briefly outlines some of the issues associated with estimating the model described in Sect. "Identification and the nature of unobserved heterogeneity". The approach we propose entails estimating a system of equations describing organizational design performance jointly with the adoption of organizational practices.

The recent literature on semi-parametric methods suggests several potential estimators for (10) and (11) (see, for example, Thompson (1989), Ichimura and Lee (1991), and Cosslett et al. (1991)). An alternative is to posit a parameterized distribution over the unobservables, which implies that estimation is feasible with GMM, or more generally, simulated method of moments (Hansen (1982), McFadden (1989), or McFadden and Ruud (1994)). Parametric approaches may be most appropriate in applications for several reasons. First, many of the applications that we might wish to consider have inherently limited samples. For example, there may be only a small number of firms within a narrowly defined industry, and the time and expense of gathering detailed data about internal organization may impose further constraints. Second, allowing for a distribution of

[28] Proposition 9 (i) does not, however, give conditions for the identification of the full joint distribution of $\Delta\chi_j^t$. It can be shown, in fact, that the variance of each $\Delta\chi_j^t$ is not identified; only $\mathrm{var}\left(\sum_{j=1}^J \chi_j^{k_j,t}\right)$ and $\sum_{j=1}^J \mathrm{var}\left(\Delta\chi_j^{k_j,t}\right)$ are identified. This remains true even under the stronger Assumption 1. On the other hand, all of the information needed to compute the selection bias is available.

[29] The restrictions imposed in (15) include both variance and covariance restrictions on the distribution of $\Delta\zeta_y^t + \Delta\nu_y^t$. However, if only the variances are known, (RPM) is not in fact a testable restriction. The hypothesis that $\zeta_{11,y_{-y}}^t + \zeta_{00,y_{-y}}^t - \zeta_{10,y_{-y}}^t + \zeta_{10,y_{-j}}^t = 0$ is not in general testable without knowledge of the covariances between the variables, even if we make a distributional assumption such as joint normality. In general, for distributions over ζ^t in suitable families (such as normal distributions), it will often be possible to satisfy restrictions such as $\mathrm{Var}\left(\zeta_{11,y_{-4}}^t + \zeta_{00,y_{-y}}^t - \zeta_{10,y_{-y}}^t + \zeta_{10,y_{-y}}^t\right) = 0$, simply by choosing the appropriate covariance restrictions on ζ^t.

unobservables with a variance-covariance matrix which is unrestricted (such as the multivariate normal) provides a parsimonious specification that still accommodates the main alternative hypotheses about unobserved heterogeneity analyzed in this paper. Thus, even if semi-parametric estimation is included in an analysis of organizational design, there may still be value to the parametric models for the purposes of summarizing the economically relevant properties of the distribution and testing hypotheses about them.

Given distributions F and G, we can specify the following set of moment equations, where we have imposed (OPT) to illustrate the cross-equation restrictions (and where v_j is a vector of appropriately specified instruments):

$$m^t(\mathrm{y}^t,\mathrm{z}^t,\mathrm{u}^t;\mathrm{a},F(\cdot),G(\cdot)) = \begin{pmatrix} \left[y_1^t - \int_{\zeta^t+v^t} y_{1^*}\left(\mathrm{z}^t,\mathrm{u}^t,\zeta^t+v^t;\mathrm{a}\right) dG\left(\zeta^t+v^t\right)\right]\cdot \mathrm{v}_1^t \\ \vdots \\ \left[y_J^t - \int_{\zeta^t+v^t} y_{J^*}\left(\mathrm{z}^t,\mathrm{u}^t,\zeta^t+\nu^t;\mathrm{a}\right) dG\left(\zeta^t+\nu^t\right)\right]\cdot \mathrm{v}_J^t \\ \left[f^t - E_{\zeta^t}\left[f\left(\mathrm{y}^t,\mathrm{z}^t,\zeta^t;\mathrm{a}\right) \mid \mathrm{y}_*^t\left(\mathrm{z}^t,\mathrm{u}^t,\zeta^t+\nu^t;\mathrm{a}\right)=\mathrm{y}^t\right]\right]\cdot \mathrm{v}_{J+1}^t \end{pmatrix} \quad (16)$$

Under the assumptions in Proposition 8 and (OPT), $E\left[m^t\left(\mathrm{y}^t,\mathrm{z}^t,\mathrm{u}^t;\mathrm{a},F(\cdot),G(\cdot)\right)\right]=0$.[30] When (RPM) is imposed, the assumptions of Proposition 9 and (OPT) suffice. In either case, we can derive the parameters describing complementarity from the estimated distributions of unobserved returns. Perhaps the most important efficiency advantage of (16) arises from the fact that organizational design practice data are often "clustered" (which we showed in Sect. "Unobserved heterogeneity and reduced-form tests for complementarity" could be a consequence of complementarity). The cross-equation restrictions in (16) allow us to estimate the distribution of incremental returns to a system, $\Delta\zeta_{\mathrm{k}}^{\mathrm{k}}$, from the adoption equations.

Under (OPT), (NI), and the (RPM) restriction, complementarity can be tested even if some combinations of practices are never observed, based entirely on the estimates of the incremental returns to each practice from the adoption equations. For example, in the simple two-choice example, there are two estimable parameters, $\theta_{00}-\theta_{01}$ and $\theta_{10}-\theta_{11}$, and four systems. Since one of the systems is normalized to determine location, we are left with three parameters $E\left[\Delta\zeta_{01}^t+\Delta\nu_{01}^t\right], E\left[\Delta\zeta_{10}^t+\Delta\nu_{10}^t\right]$, and $E\left[\Delta\zeta_{11}^t+\Delta\nu_{11}^t\right]$. Thus, $\theta_{00}-\theta_{01}$ and $\theta_{10}-\theta_{11}$ are overidentified.

The hypothesis test for complementarity involves a one-sided inequality restrictions (e.g., $\kappa\geq 0$) under the null hypothesis. In a simple 2-choice variable model, only one inequality constraint is implied by the model ($\kappa\equiv\theta_{00}+\theta_{11}-\theta_{01}-\theta_{10}$), and so the test will be a simple onetailed t-test. However, even in this case, one needs to distinguish between two alternative null hypotheses: $\kappa\geq 0$ and $\kappa\leq 0$. Of course, the latter hypothesis is the more conservative one; only a statistically significant positive coefficient on the test statistic according to a one-tailed t-test will be evidence in favor of complementarity. [31]

Testing theories about the adoption of organizational design practices

While most of our analysis has been centered on testing of complementarity, the model suggests a set of natural cross-equation restrictions regarding adoption behavior. Hypotheses about the nature of the adoption process are particularly salient since economists and other social scientists often disagree about the nature of this process and its implications for policy and strategy. At one extreme, neoclassical economics assumes that production decisions are chosen to maximize firm profits taking as given a vector of observable input prices. While transaction costs and adjustment costs have been incorporated into the literature over time, the assumption in the economics literature is still that firms are doing as well as possible subject to constraints.

In contrast, researchers in organizational behavior or who focus on strategy as an evolutionary process hold a very different model of the relationship between organizational design choice and performance. For example, the "garbage can" theory of organizations maintains that agents may be systematically misinformed about the costs and benefits of different practices within their own organization (Cohen et al. 1972). More precisely, evolutionary economists and organizational ecologists suggest that firms change only slowly and not necessarily systematically; rather, a process of "selection" eliminates firms which are poorly adapted to the current environment (Nelson and Winter 1982; Hannan and Freeman 1989). More specifically, recent research on interdependencies has explicitly tied such cognitive limitations on firm decision-making to the presence of multiple

[30] To interpret (16), note that the last equation is simply the difference between observed productivity and the expected value of productivity given observables. We will refer to this equation as the "productivity equation." In contrast, we call the first J equations the "adoption equations." For each choice, the moment condition is equal to the difference between the observed value of the choice, and the expected value of the choice given the observables and the parameters.

[31] When there are more than two choices, more subtle issues arise. In particular, pairwise complementarity may involve multiple inequality restrictions if one allows for the degree of complementarity to be dependent on other factors. If there are J practices, each restriction imposes pairwise complementarity between the two practices of interest for each of the other $J-2$ choices (resulting in $2^{(J-2)}$ linear inequality restrictions for each pairwise test). As derived by Gouriéroux et al. (1981), Wolak (1989, 1991) and Kodde and Palm (1986), the appropriate test statistic must account for the presence of multiple slack restrictions and the null hypothesis will be distributed according to a weighted sum of chi-squared distributions. Restricting the production function to be composed only of pairwise interaction effects allows the test for complementarity to be composed of a single linear inequality restriction.

(and difficult-toevaluate) interdependencies in organizational design and strategy choice (Levinthal 1997; Rivkin and Siggelkow 2003).

Despite the sharp differences between these perspectives, our empirical model is rich enough to allow for all of these possibilities. Let us briefly consider the conditions under which testable hypotheses can be formulated. We are particularly interested in whether the adoption patterns are consistent with the interaction effects estimated in the productivity equations.

In the context of our model, the set of testable restrictions about optimality is directly linked to the nature of the unobserved heterogeneity. Allowing for unobserved returns to practices outside the production function (in our model, ω) necessarily prevents us from drawing conclusions about the optimality of the propensity of firms to adopt an individual practice: an organization might simply have low unobserved returns to that practice. However, in the random practice model (RPM), there should be a single complementarity parameter for all firms. Thus, it is possible to identify differences in the (unique) complementarity parameter estimated in the productivity equation versus the (appropriately scaled) complementarity parameter estimated in the adoption equations.[32] In particular, if (NI) holds (and all of the productive interaction effects are captured in the production function), we can test whether the firm over- or underexploits the interactions between practices. For example, decentralized decision-making can lead to decisions that fail to incorporate all of the externalities of practice adoption decisions. A finding of complementarity in the production function, but a negligible interaction effect in the adoption equations, would support the hypothesis that organizations fail to account for interaction effects.

Unfortunately, if the (RPM) model is not supported in an application, it will be more difficult to test theories about adoption. By allowing for unobserved interaction effects in the adoption equations that differ from those in the productivity equations (in our model, the variables ν), we will never be able to distinguish whether firms' patterns of adoption violate optimality, or are instead simply responding to unobservables. At best, we will be able to compare the distributions of interaction effects from the adoption and productivity equations.[33]

[32] Observe that the scaling of the complementarity parameter from the adoption equation can be normalized with reference to the estimated incremental returns to an individual practice.

[33] Even if (RPM) fails, we can still draw some inferences about the adoption process if there are variables (such as x in our model) which theory suggests should affect productivity in use, but not adoption.

Issues for data gathering and survey design

Our results have several implications about the kinds of data that will be useful for testing theories about complementarities. Our analysis indicates that when the restrictions of the "random practice model" are satisfied, there will a single parameter of complementarity between any two practices for all firms, and we can test and exploit cross-equation restrictions between adoption and productivity equations. For this reason, we believe that the framework is most suitable when applied with a narrow industry with well-defined practice adoption choices that are the same for all firms in an industry.

As well, a key requirement for testing for complementarity under the RPM is measuring choice-specific instrumental variables, which are closely associated with the adoption of individual practices but independent of the adoption of other practices or realized performance (except through the impact on adoption behavior itself). These measures are essential for satisfying the assumptions (such as EXCL or Assumption 2) required to implement many of the procedures discussed in this paper. Since it is necessary to observe variables affecting adoption decisions but not productivity, the choice of instruments and performance measures must go hand in hand. In particular, to manage the problems created by the unobserved heterogeneity, it is useful to find the narrowest possible measure of productivity which still incorporates all of the interactions between endogenous variables. This makes it easier to find instruments (w) which represent costs to the organization that do not interact directly with productivity. Indeed, if the performance measure is suitably narrow so that unobserved heterogeneity is due to ω and not χ, OLS will yield unbiased results.

However, it is important to emphasize that, particularly in the context of a large data-gathering project (e.g., a multi-year multi-site study), it would be extremely useful to gather both choice-specific and system-specific instrumental variables. A dataset which includes both types of measures would be required in order to estimate the RPM (under Assumption 2) and compare those results to the results of an RSM estimation (under Assumption 1).

Further, to the extent that observed choice variables are positively correlated and unobserved heterogeneity is important, testing for complementarity relies on the adoption equations, and so it is crucial to understand the adoption process. The approach suggested here will be most powerful in applications where adoption is relatively systematic and can be at least partially explained by observables (whether or not adoption is "profit-maximizing" in a strict sense). If the adoption process is too noisy, very little will be learned from estimating adoption equations. For this reason, it will be difficult to analyze applications where

there is rapid diffusion of organizational practices and the researcher only observes a cross-section. In particular, difficulties will arise if firms are adopting sets of practices together, without fully understanding their interactions, and if the choice of which firms adopt is determined by factors such as central management's taste for management fads.[34] For survey design, focusing interview and survey questions on the factors that enter the adoption process are critical for justifying the econometric approach used to test for complementarity.

Of course, if adoption is completely unrelated to productivity (as predicted by some of the theories from sociology described above), it may be possible to treat the determination of organizational design as exogenous even in cross-sectional studies. However, it will be difficult to justify such an exogeneity assumption in practice. As such, for any cross-sectional study, it is important to collect data that allows for a clear test of assumptions made about the exogeneity (or not) of organizational design choice. In contrast, if it is possible to observe performance both before and after adoption, it may be feasible to implement a "differences-in-differences" approach, which compares the rate of improvement in performance among adoptors to the rate of improvement in performance among those who have not yet adopted the full set of potentially complementary practices (see Athey and Stern (2002) for a fuller discussion and implementation of this approach).

Finally, our analysis highlights the central role of unobserved returns to different practices, and their statistical interrelationships, on the biases associated with a variety of simple econometric procedures. Thus, even if systematic quantitative evidence is impossible to gather, qualitative information about the factors which affect the adoption and productivity of practices, and their covariation, can play a central role in distinguishing between alternative theories and interpreting the results of procedures such as (CORR) and (PROD). This is particularly relevant as many of the datasets and quantitative investigations undertaken in this area are complemented by detailed knowledge by researchers of the circumstances of adoption (see Ichniowski and Shaw (2003) for a more detailed discussion of the benefits and costs of "insider econometrics.").

In summary, our proposed method can be most fruitfully applied in scenarios where there is a precise and suitably narrow productivity measure available, the adoption process is systematic, firms face the same basic production technology, and it is possible to observe information about the costs and benefits to adoption, particularly costs and benefits which do not interact with productivity directly. Potential examples include customer service organizations or service industries, such as banks or retail outlets, where many outlets serve a variety of customer types and are located in a variety of labor market and regulatory environments.

Conclusions

Understanding the sources of inter-firm heterogeneity, and the nature and importance of complementarities between practices, is important for both business and public policy. This paper highlights many of the difficulties that arise in trying to disentangle different hypotheses about the causes of positive correlation between organizational choice variables. In particular, we show that the approaches which have been most commonly used in the literature can yield misleading results when we allow for complementarities between choice variables as well as unobserved factors which affect the marginal costs and benefits of each individual choice.

The empirical framework we propose is tailored to disentangle the different forces behind the observation of "clustered" organizational design practices. Using the simultaneous equations systems proposed in this paper, we can, in principle, distinguish between two competing assumptions about the nature of unobserved heterogeneity, test for the cross-equation restrictions associated with static optimization, and, most importantly, provide a consistent test statistic for complementarity. As well, this system of equations approach can provide substantial increases in precision, particularly important when the sample sizes associated with many applications are small and there is a tendency towards clustering among the dependent variables.

In this paper, we have developed a baseline framework. Several interesting theoretical issues remain to be explored. For example, we wish to study more carefully the issues associated with aggregating organizational design variables, in particular the use of "indices" to describe the adoption of a set of organizational design practices (as in ISP). Further, this paper has not addressed the issues associated with the dynamics of the diffusion and adoption of organizational design practices. As well, a crucial and related step is the implementation of these techniques in real-world data sets, which we are pursuing in several related projects.

[34] For example, during the Internet boom, several studies emphasized potential complementarity between organizational practices and the adoption of Internet-related technologies; it is likely that at least some results were contaminated by the type of unobserved heterogeneity highlighted by our model.

Appendix A

Proof of Proposition 3 (PROD) Abstract away from x and w. It is clear from Fig. 1 that $E\left[\chi_1^{1,t}|\chi^{1,t}+\boldsymbol{\omega}^{1,t}\in R_{11}\right]$ $\leq E\left[\chi_1^{1,t}|\chi_1^{1,t}+\omega_1^{1,t}\geq\theta_{00}-\theta_{10}\right]$ $=E\left[\chi_1^{1,t}|\chi^{1,t}+\boldsymbol{\omega}^{1,t}\in R_{10}\right]$ since the random variables are independent and since $\kappa\geq 0$ (notice that R_{11} includes regions which are strictly less than R_{10}) The same logic implies that $E\left[\chi_2^{1,t}\mid\chi^{1,t}+\boldsymbol{\omega}^{1,t}\in R_{11}\right]\leq E\left[\chi_2^{1,t}\mid\chi^{1,t}+\boldsymbol{\omega}^{1,t}\in R_{01}\right]$, yielding the desired result.

(CORR) By Proposition $1, \mathrm{y}^*\left(\chi^{1,t},\boldsymbol{\omega}^{1,t}\mid \mathrm{x}^t,\mathrm{w}^t\right)$ is nondecreasing in $\chi^{1,t}$ and $\boldsymbol{\omega}^{1,t}$, and $\left(\chi^{1,t},\omega^{1,t}\right)$ are independent. Then, we use the fact that a vector of nondecreasing functions of independent random variables is positively correlated. □

Proof of Proposition 4 (PROD) When $\kappa=0,\theta_{00}-\theta_{10}=\theta_{01}-\theta_{11}$. Given W0, $R_{11}=\{(\chi_1,\chi_2)\mid(\chi_1,\chi_2)\geq(\theta_{00}-\theta_{10},\theta_{00}-\theta_{01})\}$, while $R_{10}=\{(\chi_1,\chi_2)\mid\chi_1\geq\theta_{00}-\theta_{10},\chi_2\leq\theta_{00}-\theta_{01}\}$. We now apply a theorem from Athey (2002), which states that if a set A is larger than B in the "strong set order," and the random variables are affiliated, then $E\left[g\left(\chi_1,\chi_2\right)\mid\left(\chi_1,\chi_2\right)\in A\right]\geq E\left[g\left(\chi_1,\chi_2\right)\mid\left(\chi_1,\chi_2\right)\in B\right]$ for any nondecreasing function g. Since $g\left(\chi_1,\chi_2\right)=\chi_1$ is nondecreasing, and R_{11} is larger than R_{10} in the strong set order, the theorem applies to yield the result that

$$E\left[\chi_1\mid\left(\chi_1,\chi_2\right)\in R_{11}\right]\geq E\left[\chi_1\mid\left(\chi_1,\chi_2\right)\in R_{10}\right].$$

Similar arguments applied to χ_2 lead to the result that $E\left[\hat{\kappa}^{OLS}\right]\geq 0$.

(CORR) The result follows because nondecreasing functions of affiliated random variables are affiliated. □

Proof of Proposition 5 Part (a) follows from the proof of Proposition 3, while part (b) follows from the proof of Proposition 4. □

Proof of Proposition 9 Suppress z for simplicity. Normalize w^1 so that all of the coefficients δ_j^1 are positive (ORD). For notational simplicity, focus on the case where $i=1$ and $j=2$, and fix $\mathrm{y}-12$ and $\mathrm{w}^{0,t}$. Start with part (ii). Under Assumption 2, letting $i=1$ and $j=2$ for notational simplicity, the following function is identified from the observables:

$$\begin{aligned}&\lim_{\left(-w_2^{1,t},2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\mathrm{y}^t=1,0,\mathrm{y}_{-12}\mid\mathrm{w}^{0,t},\mathrm{w}^{1,t}\right)\\&=\lim_{\left(-w_2^{1,t},2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\theta_{1,0,\mathrm{y}_{-12}}-\theta_{0,0,\mathrm{y}_{-12}}+\Delta\chi_1^t+\Delta\omega_1^t\geq\right.\\&\quad\left.\delta_1^0w_1^{0,t}-\delta_1^1w_1^{1,t}\mid\mathrm{w}^{0,t},\mathrm{w}^{1,t}\right)\end{aligned}\tag{17}$$

Normalize $\delta_j^1=1$ for each j. Then, varying $w_1^{1,t}$ throughout its support determines the distribution of $\theta_{1,0,y_{-12}}-\theta_{0,0,y_{-12}}+\Delta\chi_1^{1,t}+\Delta\omega_1^{1,t}$. Similarly, the following function is identified from the observables:

$$\begin{aligned}&\lim_{\left(w_2^{1,t},2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\mathrm{y}^t=1,1,\mathrm{y}_{-12}\mid\mathrm{w}^{0,t},\mathrm{w}^{1,t}\right)\\&=\lim_{\left(w_2^{1,t},2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\theta_{1,1,\mathrm{y}_{-12}}-\theta_{0,1,\mathrm{y}_{-12}}+\Delta\chi_1^t+\Delta\omega_1^t\geq\right.\\&\quad\left.\delta_1^0w_1^{0,t}-\delta_1^1w_1^{1,t}\mid\mathrm{w}^{0,t},\mathrm{w}^{1,t}\right)\end{aligned}\tag{18}$$

Then, varying $w_1^{1,t}$ throughout its support determines the distribution of $\theta_{1,1,y_{-12}}-\theta_{0,1,y_{-12}}+\Delta\chi_1^{1,t}+\Delta\omega_1^{1,t}$. Then, the difference between the two distributions is $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$. Since we normalized the scale of $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$ when we normalized δ_j^1 for each j, we have only identified the sign of $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$.

Now suppose that $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$ is non-negative. Then, the following function is identified from the observables (see Fig. 1):

$$\begin{aligned}&\lim_{\left(2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\mathbf{y}^t=1,0,\mathbf{y}_{-12}|\mathbf{w}^{0,t},\mathbf{w}^{1,t}\right)\\&=\lim_{\left(2\left(y_3-\frac{1}{2}\right)w_3^{1,t},\ldots,2\left(y_J-\frac{1}{2}\right)w_j^{1,t}\right)\to(\infty,\ldots,\infty)}\\&\quad\Pr\left(\begin{array}{l}\theta_{1,0,\mathbf{y}_{-12}}-\theta_{0,0,\mathbf{y}_{-12}}+\Delta\chi_1^t+\Delta\omega_1^t\geq\\\delta_1^0w_1^{0,t}-\delta_1^1w_1^{1,t}\\\theta_{1,1,1,\mathbf{y}_{-12}}-\theta_{1,0,\mathbf{y}_{-12}}+\Delta\chi_2^t+\Delta\omega_2^t\leq\\\delta_2^0w_2^{0,t}-\delta_2^1w_2^{1,t}\end{array}\mid\mathbf{w}^{0,t},\mathbf{w}^{1,t}\right)\end{aligned}\tag{19}$$

Maintaining the normalizations $\delta_j^1=1$ for $j=1,..,J$, varying $w_1^{1,t}$ and $w_2^{1,t}$ throughout their supports identifies the joint distribution of $(\Delta\chi_1^t+\Delta\omega_1^t,\Delta\chi_2^t+\Delta\omega_2^t)$ (marginalized over the unobserved returns to the other practices) up to location, given that we cannot distinguish changes in $\theta_{1,0,y_{-12}}-\theta_{0,0,y_{-12}}$ and $\theta_{1,1,y_{-12}}-\theta_{1,0,y_{-12}}$ from additive shifts in the distribution of $(\Delta\chi_1^t+\Delta\omega_1^t,\Delta\chi_2^t+\Delta\omega_2^t)$. Normalizing the location of $(\Delta\chi_1^t+\Delta\omega_1^t,\Delta\chi_2^t+\Delta\omega_2^t)$ restores identification.

Now, consider part (i). Fix y. Then the following function is identified from the observables for each real number c, using the notation $\zeta_{\mathrm{y}}^t = \sum_{j=1}^J \chi_j^{y_j,t}$:

$$= \lim_{\left(2(y_1-1/2)w_1^{1,t},\ldots,2(y_J-1/2)w_j^{1,t}\right)\to(\infty,\ldots,\infty)} \Pr\left(f^t \le c \mid \mathrm{w}^{0,t}, \mathrm{w}^{1,t}\right)$$
$$\lim_{\left(2(y_1-1/2)w_1^{1,t},\ldots,2(y_J-1/2)w_j^{1,t}\right)\to(\infty,\ldots,\infty)} \Pr\left(\theta_{\mathrm{y}} + \zeta_{\mathrm{y}}^t \le c \mid \mathrm{w}^{0,t}, \mathrm{w}^{1,t}\right)$$

Thus, the marginal distribution of $\theta_{\mathrm{y}} + \zeta_{\mathrm{y}}^t$ is identified up to location for each y. If we normalize $\chi_j^{y_j,t}$ to have mean zero for each j and y_j, then θ_{y} is identified for each y. Notice that the location normalization does not affect the complementarity parameters $\kappa_{ij}\left(\mathrm{y}_{-ij}\right)$, since these are calculated as differences (in general, we would take differences of the means of $\theta_{\mathrm{y}} + \sum_{j=1}^J \chi_j^{y_j,t}$ for the appropriate y 's, and shifts in the distributions of $\chi_j^{y_j,t}$ would cancel out).

To show that $\mathrm{cov}\left(\chi_i^{1,t}, \chi_j^{1,t}\right)$ is identified, let $\psi(\mathrm{y}) = \mathrm{var}\left(\sum_{j=1}^J \chi_j^{y_j,t}\right)$, and note that we have just established identification of $\psi(\mathrm{y})$ for each y. Expanding, we have $\mathrm{var}(\psi(\mathrm{y})) = \sum_{j=1}^J \mathrm{var}\left(\chi_j^{y_j,t}\right) + 2\sum_{i=1}^J \sum_{j=1}^{i-1} \mathrm{cov}\left(\chi_i^{y_i,t}, \chi_j^{y_j,t}\right)$. But $\mathrm{cov}\left(\Delta\chi_j^t, \Delta\chi_k^t\right) = \mathrm{cov}\left(\chi_i^{1,t}, \chi_j^{1,t}\right) + \mathrm{cov}\left(\chi_i^{0,t}, \chi_j^{0,t}\right) - \mathrm{cov}\left(\chi_i^{1,t}, \chi_j^{0,t}\right) - \mathrm{cov}\left(\chi_i^{0,t}, \chi_j^{1,t}\right)$. It then follows that $2\mathrm{cov}\left(\Delta\chi_i^t, \Delta\chi_j^t\right) = \psi(0,0;\mathbf{y}_{-ij}) + \psi(1,1;\mathbf{y}_{-ij}) - \psi(0,1;\mathbf{y}_{-ij}) - \psi(1,0;\mathbf{y}_{-ij})$ □

Appendix B

This brief appendix provides a specific example where comparing results from two LATE estimators does not provide a consistent estimate of the complementarity parameter. Consider the case where two practices are in fact complements, as depicted in Fig. 4. When $w_2 = 0$, in response to an increase in w_1, the optimal response for firms with unobservables in region B is to switch from (0, 1) to (1, 1), while the optimum in regions F and G switches from (0, 0) to (1, 1), and the optimum in regions J, K, O, and P switches from (0, 0) to (1, 0). When $w_2 = 1$, in response to an increase in w_1, regions B, E, F, I, and J switch from (0, 1) to (1, 1), regions N and O switch from (0, 0) to (1, 1), and region P switches from (0, 0) to (1, 0). Thus, in the presence of complementarity, the LATE estimators for the cases where $w_2 = 0$ and $w_2 = 1$ are:

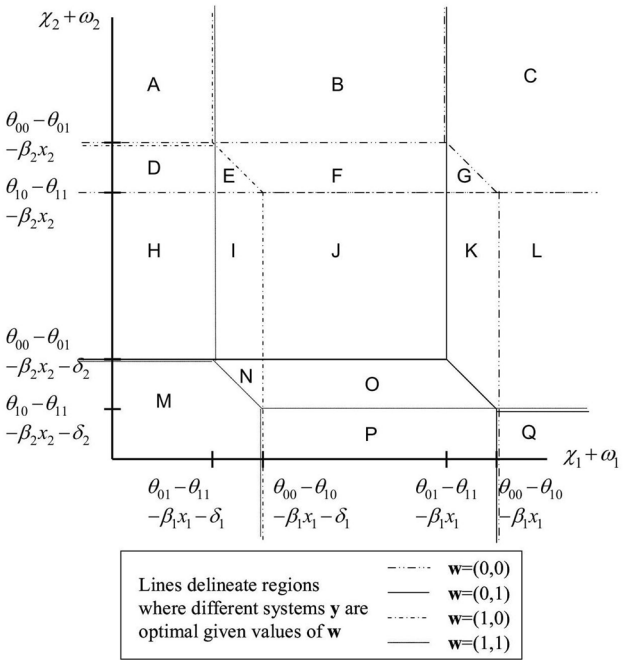


Fig. 4 Regions of adoption for different values of the instrumental variables: complementary practices

$$\psi_0 = \frac{1}{\Pr(B\cup F\cup G\cup J\cup K\cup O\cup P)} \times \left\{ \begin{array}{l} (\theta_{11}-\theta_{01}+E[\chi_1 \mid B])\Pr(B) \\ +(\theta_{10}-\theta_{00}+E[\chi_1 \mid J\cup K\cup O\cup P])\Pr(J\cup K\cup O\cup P) \\ +(\theta_{11}-\theta_{00}+E[\chi_1+\chi_2 \mid F\cup G])\Pr(F\cup G) \end{array} \right\}$$

$$\psi_1 = \frac{1}{\Pr(B\cup E\cup F\cup I\cup J\cup N\cup O\cup P)} \times \left\{ \begin{array}{l} (\theta_{11}-\theta_{01}+E[\chi_1 \mid B\cup E\cup F\cup I\cup J])\Pr(B\cup E\cup F\cup I\cup J) \\ +(\theta_{10}-\theta_{00}+E[\chi_1 \mid P])\Pr(P) \\ +(\theta_{11}-\theta_{00}+E[\chi_1+\chi_2 \mid N\cup O])\Pr(N\cup O) \end{array} \right\}$$

These two expressions cannot be compared without distributional assumptions about the probability of different regions. Suppose for the moment that the unobserved returns are independent and uniformly distributed over the relevant region, so that the denominators in the two expressions are equal, and further $\Pr(N\cup O) = \Pr(F\cup G)$ and $\Pr(E\cup F\cup I) = \Pr(O\cup K)$. Then, we have

$$\begin{aligned} \psi_1 - \psi_0 = & \frac{1}{\Pr(B\cup E\cup F\cup I\cup J\cup N\cup O\cup P)} \\ & \times \{(\theta_{11}-\theta_{01}-(\theta_{10}-\theta_{00}))\Pr(J) + 0\cdot\Pr(B\cup P) \\ & +(E[\chi_1 \mid E\cup F\cup I]-E[\chi_1 \mid K\cup O])\Pr(E\cup F\cup I) \\ & +(E[\chi_1+\chi_2 \mid N\cup O]-E[\chi_1+\chi_2 \mid F\cup G])\Pr(N\cup O)\} \end{aligned}$$

The right-hand side of this expression is the weighted average of several terms. The first term in brackets is the true complementarity parameter, weighted by the probability of the region J. Region J is the region that experiences all four combination of practices as a result of changes in w; it is analogous to region E in Fig. 3. The second term is 0 . The third term is the difference between the expected unobserved

returns to y_1 in regions E, F, and I versus regions K and O. If the unobserved returns are independent across practices, this is negative; intuitively, a less productive selection of firms is induced to adopt y_1 when $w_2 = 1$. Similarly, the fourth term is negative as well. Thus, when the unobserved returns are uniformly distributed, the difference between the LATE estimators will typically be less than the true complementarity parameter. This is due to selection bias, since the set of firms that adopts y_1 when $w_2 = 1$ is less productive than when $w_2 = 0$. Indeed, it is possible to construct simulations (available on request) where the complementarity parameter is positive but moderate in size, the selection bias is substantial, yielding a negative estimate of $\psi_1 - \psi_0$ even though the practices are in fact complements

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